Methods used for Household Food and Drink Waste in the UK 2022

METHODOLOGICAL REPORT

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About WRAP

WRAP is a climate action NGO working around the globe to tackle the causes of the climate crisis and give the planet a sustainable future.

Our core purpose is to help you tackle climate change and protect our planet by changing the way things are produced, consumed, and disposed of.

This document provides a detailed account of the methodology used in the PFU005 project. This project quantifies the total amount of household food waste in the UK as well as the types of food wasted, the reasons food is wasted and the impact of wasted food (financial and environmental).

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1.0 Introduction

This document contains details of the research methods and analysis used to obtain the results for Household Food and Drink Waste in the United Kingdom 2022. This replaces the document previously published in 2023 (no longer available), which looked at the research methods and analysis used to obtain results for Household Food and drink Waste in the United Kingdom 2021/22. The reasons for this update were:

- Food waste was changing rapidly from 2021 to 2022, so presenting data for 2022 will be more useful into the future.
- Some inconsistencies in how the amount of food waste was calculated were found in the calculations for 2021 data. It was decided to recalculate the data and, rather than articulate data for 2021/22, present the data for 2022 for the reason above.

1.1 Background and history

WRAP have measured household food waste in the UK since 2007. This includes trend data, which monitors how the amount of household food waste has changed over time. WRAP research estimates the total amount of local authority collected household food waste[1], such studies have been published by WRAP at least every 2-3 years allowing trends over time to be analysed. More detailed food waste studies have been conducted in both 2007 and 2012[2] which includes a detailed waste composition analysis of food waste, diaries research as well as a local authority collected household waste study. The inclusion of all of these elements allows the total amount of household food waste in the UK to be estimated by including data on food waste which is home composted or disposed of via the sewer. Additionally, details of exactly what food is wasted, reasons for discarding and the impact of household food waste (environmental and financial) can be gained from these more detailed pieces of research.

The 2012 household food waste results were restated in 2015[3], using the data collected in 2012. The primary difference in 2015 compared to 2012 is the categorisation of food as edible and inedible instead of unavoidable, possibly avoidable, and avoidable. Additionally, food fed to pets was omitted from counting towards household food waste. These 2015 updates to the methodology have, therefore, also been applied in this updated research.

Given that 2012 was the last time either a detailed waste composition analysis and food waste diaries study was conducted, an update of household food waste data was required. Household

[1] Local authority collected household food waste in the UK 2022, WRAP 2025, Via: https://www.wrap.ngo/resources/report/local-authority-collected-household-food-waste-uk-2022
[2] Household Food and Drink Waste in the United Kingdom 2012, WRAP, Accessed in: August 2023, Via: Household Food and Drink Waste in the United Kingdom 2012 | WRAP
[3] Courtauld 2025 baseline and restated household food waste figures, WRAP, Accessed in: August 2023, Via: Courtauld 2025 baseline and restated household food waste figures | WRAP

food waste data is needed for several things, such as to assess against Courtauld and SDG (Sustainable Development Goal) targets, to understand precisely what types of food are wasted, why it is wasted and the impact of food waste (financial and environmental). Therefore, this research aims to update this data on household food waste in the UK.

1.2 Summary of research methods used

The research is comprised of the following pieces of primary research:

1. Detailed waste compositional analysis 21/22: This study classifies and quantifies food and drink waste from local authority collected waste streams.
2. Kitchen and sewer diaries 22/23: The diaries collect information on both the amounts and types of food wasted as well as the reasons for discarding.
3. Local authority collected household food waste in the UK 2022: The analysis gives an estimate of the total amount of household food and drink waste in local authority collected waste streams.

Each piece of primary research is outlined in more detail in section 2.1, which also outlines where detailed methodologies can be found for each element of the research. Table 1 details how each of these primary pieces of research are used to reach different research outcomes (Total amount of food waste, Types of food wasted and reasons for disposal).

Table 1: The primary data used to reach each research outcome by disposal route.

Disposal route	Total amount of food wasted	Types of food wasted	Reasons for discarding
Local authority collected	Local authority collected household food waste in the UK 2022	Detailed waste composition analysis	General kitchen food waste diaries
Sewer	Sewer diaries	Sewer diaries	Sewer diaries
Home composting	General kitchen food waste diaries	General kitchen food waste diaries	General kitchen food waste diaries

1.3 Report structure

The reasons that each element of the research (as outlined in Table 1) has been used in the way it has is discussed in Chapter 2.0. The methodology used to gather the primary data for the Waste composition analysis and diaries is described in detail in Chapters 3.0 and 4.0. Local authority collected household food waste in the UK 2022 is presented in a separate report[4].

Calculations are performed on this data (Chapter 5.0) and weightings (Chapter 6.0) applied to enable an understanding of the total food wasted in the UK and the types of food wasted. These results indicated that the diaries data is suspected of underestimating food waste which is explored in detail in Chapter 7.0. Further analysis of the UK food waste totals was also undertaken to understand reasons for discarding, the impact of food waste (financial and environmental) and the proportion of food purchased wasted as outlined in Chapter 8.0. An overview of the different elements of data and calculations applied is shown is shown at the start of Chapter 5.0 in Figure 4.

1.4 Food waste definitions

In this report, the same definitions of food waste as in the household food waste statistics restated for 2007 to 2015 have been utilised[5].

The terminology used is with the edibility classification system is as follows:

- Food waste: Food and the inedible parts of food removed from the food supply chain (or household) to be recovered or disposed of (including - composted, anaerobic digestion, incineration, disposal to sewer or landfill). This definition excludes waste prevention activities, namely redistribution for human consumption, or diverted to feed animals[6].
- Wasted food (also referred to as edible parts): products (or parts of products) intended for human consumption. This includes material which is still suitable for consumption when it is disposed of (i.e. would be regarded as 'edible') and that which may no longer be suitable for consumption at the point of discarding (e.g. for example due to it passing a 'use by' date or being spoiled). It excludes inedible parts.
- Inedible parts: Components associated with a food that are not intended for human consumption. Examples of inedible parts associated with food could include bones, rinds, and pits/stones. "Inedible parts" do not include packaging. What is considered inedible

[4] Local authority collected household food waste in the UK 2022, WRAP 2025, Accessed n November 2024, Via https://www.wrap.ngo/resources/report/local-authority-collected-household-food-waste-uk-2022
[5] Household food waste: restated data for 2007-2015, WRAP, 2018, Accessed in August 2023, Via: Technical report templates (wrap.org.uk)
[6] The FUSIONS project, through extensive consultation, published a food waste 'definitional framework' and definition which is largely consistent with this definition, in terms of covering both food and inedible parts and the relevant destinations. See the following: FUSIONS Definitional Framework for Food Waste 2014.pdf (eu-fusions.org)

varies among users (e.g., chicken feet are consumed in some food supply chains but not others), changes over time, and is influenced by a range of variables including culture, socio-economic factors, availability, price, technological advances, international trade, and geography.

More detail on the calculations used to determine the amount of edible and inedible food is in section 5.3.

2.0 Rationale for primary research data used

This section outlines the primary research conducted and the rationale for why particular data has been used.

2.1 Primary research conducted

The research is comprised of the following pieces of primary research:

1. Detailed waste compositional analysis 21/22: This study classifies and quantifies food and drink waste from 1,807 households. This gives detailed information on the types of food wasted. Additionally, households completed a face-to-face questionnaire covering sociodemographic and behavioural questions. Sampling was designed to be representative for the UK, with households in England and Wales being sampled. The detailed methodology for this part of the research is given in chapter 3.0.
2. Kitchen and sewer diaries 22/23: Information on the types of food and drink wasted was recorded in diaries by a total of 1,079 households (459 sink diaries and 620 Kitchen diaries). Two separate diaries were conducted for this research, one diary for sewer waste (sink waste) and another for kitchen waste which included food disposed of via the residual bin, separate food waste containers, home composted or fed to pets. For each instance of waste, the amount of waste, the reason for discarding and which 'waste' stream was used was recorded. The research included waste streams that are hard to measure from compositional analysis (material poured down the kitchen sink or other inlets to the sewer or home composted). Additionally, although no longer classed as food waste and not a focus of this work, the diaries also allow an understanding of the amount of food fed to animals. Each household was also surveyed before and after the diary research via a questionnaire, giving information on attitudes, stated behaviours and household characteristics. The detailed methodology for this part of the research is given in Chapter 4.0.
3. Local authority collected household food waste in the UK 2022: This research collates recent waste compositional analyses (WCA's) commissioned by local authorities. The information from these audits is used alongside waste and recycling tonnage data from WasteDataFlow – a reporting system for waste collected by local authorities in the UK. The analysis gives an estimate of the total amount of household food and drink waste in local

authority collected waste streams. See separate report: *Local authority collected household food waste in the UK 2022*[7], for more details of the methodology and report.

Timeframe of research conducted

The overall food waste estimates presented in this report are referred to as being from 2022 because of the timeframe in which different elements of the research were conducted. The majority of data has been collected in 2022, with small amounts collected at the end of 2021 and the start of 2023.

The detailed waste composition analysis was conducted at the end of 2021 and start of 2022, to provide the detail around the types of food wasted in local authority collected waste streams. The local authority collected household food waste studies, which is used to track the amount of food waste overtime utilises more general local authority collected waste composition data from July 2021 and June 2023 (estimates are pooled from a longer period to increase the number of data points). To understand the total amount of waste collected from local authorities for the local authority collected household food waste studies, data from 2022 was used from WasteDataFlow. At the time of conducting this research this was the most recent data available. The diaries were then conducted at the end of 2022 and start of 2023, with one wave of research in each year. Subsequently some of the data used to understand total food waste from home composting and sewer waste as well as reasons for discarding is from the start of 2023.

2.2 Rationale for research used: Food and drink waste collected by local authorities

Total amount of food wasted in the UK

In this study the local authority collected household food waste in the UK studies method was used to estimate the total amount of household food waste collected by local authorities (LAs), which is in line with the methodology used in previous years[8]. Besides the local authority collected household food waste in the UK studies method, for estimating LA collected waste other options include:

[7] Local authority collected household food waste in the UK 2022, WRAP 2025, Accessed via https://www.wrap.ngo/resources/report/local-authority-collected-household-food-waste-uk-2022

[8] WRAP, Methods used for Household Food and Drink Waste in the UK 2012, 2013, Accessed via: Household Food and Drink Waste in the United Kingdom 2012 | WRAP

- Scaling the information from the detailed waste compositional analysis: the average amount of food and drink waste per household in these waste streams can be calculated and scaled up to the total number of households in the UK.
- Scaling the average amount of food and drink waste per household in the kitchen diaries associated with these waste streams to the whole UK population.

The local authority collected household food waste in the UK studies method has a number of advantages for calculating the total amount of food and drink waste:

1. It uses a larger number of compositional studies already performed by local authorities to derive waste estimates. This means that the effective sample size of the study is much higher than either of the other two methods.
2. It uses information in WasteDataFlow in combination with waste composition data, thus meaning that the estimates are partially based on a census of the total amount of material in relevant waste streams. For streams where all the waste is likely to be food and drink (i.e. separate food waste collections), the information is not subject to sampling errors – again the uncertainty in the results is much lower.
3. Another strength of the method is that general waste audits undertaken by local authorities usually have less contact with the households that they are sampling from. Specifically waste audits from local authorities do not typically include a questionnaire before a waste audit, as is the case with the detailed waste compositional analysis and the kitchen diaries. This means that the degree to which households change their behaviour as a result of any contact is likely to be smaller which may result in lower participation bias.
4. Finally, because the data does not need to be linked to households, it is easier for the waste composition data to include flats and other shared properties, allowing the estimates to be more representative of the UK population of all households.

A potential disadvantage of using the local authority collected household food waste studies method is that it relies on the accuracy of data submitted on WasteDataFlow by local authorities. WasteDataFlow is a reporting system for waste collected by local authorities in the UK, in which local authorities have to report all their waste data into. Although WasteDataFlow is a well-established database of local authority collected waste and checks are undertaken on data submitted it may contain some inaccuracies. For example, waste data submitted as household waste could mistakenly include waste from schools or small businesses, resulting in an overestimate of household food waste.

On the other hand, diary methods suffer from underestimation of the amount of food and drink waste. For the food element of the local authority collected waste streams (i.e. omitting food and drink waste disposed of to the sewer and home composted), a comparison between diary research and detailed composition analysis was performed on data from WRAP for 2007 and

2012/13 (Quested et al. 2020[9]). This indicated that estimates from the diary were around 40% lower than compositional analysis in 2007 and 30% lower in 2012/13. The results from this study are broadly in line with this finding, showing a 42% difference between the WCA and diary as discussed in Chapter 7.0, although this reduces to 34% when you take into account the 8% difference in food waste tonnages between 2021 and 2022. Quested et al. (2020) concluded that a) diaries are useful for obtaining approximate estimates of HHFW and detailed information on what, why, and where food is discarded, but diaries alone are not suitable for tracking HHFW over time or evaluating interventions designed to reduce the amount of HHFW (without substantial further research).

For the above reasons, information from Local authority collected household food waste in the UK 2022 was used to provide the total amount of food and drink waste collected by local authorities from UK households.

Types of food wasted

The detailed waste composition analysis is used to understand the types of food wasted in local authority collected waste streams. Although the detailed waste compositional analysis suffers from some drawbacks in obtaining the total amount of food and drink waste, these drawbacks are less important when being used to determine the proportion of different types of food and drink thrown away. In detailed waste composition analysis, the precision of data collected creates logistical challenges and increases the required time for sampling. However, a sample of 1807 households used in this study still provides a large enough sample size to understand the types of food wasted by UK households. As a waste composition analysis is a sample of the population, random sampling errors are inevitable, however these are accounted for at the nation level in the confidence intervals presented in section 9.0. Systematic errors in sampling can then occur if the households sampled are not representative of the population. Therefore, weightings have been applied to the waste composition data to account this (see section 6.0).

Why food is wasted?

Despite the shortcomings of diary-based research, the kitchen diaries are the only reliable source of information on the reasons why food is thrown away. As in 2012, the kitchen diaries will be used to estimate reasons for discarding for all waste streams in the research. The diaries enable the reasons for discarding for each food item wasted to be specified, the diaries include a question on why something was thrown away, what the item is and the quantity. Therefore, reasons for discarding by food category can be determined. Using a questionnaire would only allow a more general assessment of why food is thrown away or it would be reliant on someone thinking back to the last time they threw something away. Subsequently, the data collected from a

[9] Quested, T. E., Palmer, G., Moreno, L. C., McDermott, C., & Schumacher, K. (2020). Comparing diaries and waste compositional analysis for measuring food waste in the home. Journal of Cleaner Production, 262, 121263.

questionnaire would be less reliable and wouldn't give as much detail on the reasons for discarding for different food categories.

2.3 Rationale for research: sewer and home composted food and drink

This section outlines why diaries research has been used for both sewer and home composted waste. It should be noted that separate diaries have been used for quantifying sewer and home composted food waste. As in the restated household food waste for 2007-2015 report[10], food fed to animals is no longer considered food waste. Therefore, although diaries record this data this is not a focus of this report and so is not discussed specifically in this section.

Total amount of food wasted in the UK

Data for food and drink disposed of via the sewer has been collected via diaries focusing only on sewer waste. This approach is aligned with the methodology used in the *Down the Drain*[11] study, which also utilised a separate diary for sewer waste only. A focused diary approach which looked only at sewer waste intended to avoid the need for participants to report details on multiple disposal routes which should mean that there is less of an underestimation. This was an issue identified with the diaries conducted in the 2012 equivalent study, in which diaries asked participants to report on multiple disposal routes.

Data for the amount of food waste home composted utilises a diary method which is the only feasible method for obtaining information on the amount of food and drink waste that is home composted. Other methods have been considered (e.g. compositional analysis of a compost heap), but major problems exist with such methods (e.g. in the example above, knowing when food was placed on the compost heap).

Diaries, as discussed above, suffer with issues with underestimation, although it is unclear the level of which this exists sewer waste and home composting as there are few (if any) comparable studies. The degree of underestimation may be substantially less than for local authority collected waste: if diaries influence behaviour, then there is the possibility that material is diverted towards disposal routes that are seen as socially desirable (such as home composting), which could counteract any incompleteness of diary entries.

[10] Household food waste: restated data for 2007-2015, WRAP, 2018, Accessed in August 2023, Via: Technical report templates (wrap.org.uk)
[11] WRAP, 2009, Down the Drain (WRAP Project EVA063). Report prepared by WRAP, Banbury.

Types of food waste

Diaries are used to understand the types of food wasted for both sewer and home composted food waste for similar reasons mentioned above. The predominant reason being that there are not suitable alternative methods for gathering data on the type of food wasted via these disposal routes.

Why food is wasted?

Diaries are again used the understand why food is wasted for sewer and home composted waste. The reasons for this are the same as those given in the 'Why food is wasted?' paragraph of section 2.2.

3.0 Methods for detailed waste compositional analysis

This section of report details the methodology of the detailed waste composition analysis component of this project. A methodology is outlined for the sampling process, the questionnaire conducted and details around the waste composition analysis.

3.1 Overview

The aim of the waste composition analysis was to classify and quantify food and drink waste from local authority collected waste streams. In addition, the project aimed to provide detailed information on the demographics and behaviours of each household sampled in the research.

To achieve these aims a detailed food waste specific compositional analysis on the waste streams of a sample of households, representative of the UK was undertaken. In addition, a questionnaire related to food, food waste, demographic information and behaviours was conducted for the same households prior to the waste compositional analysis (WCA). Data from this questionnaire will be used by WRAP in future research to understand demographic and behavioural drivers of household food waste. For example, the questionnaire will allow an exploration of whether factors such as age, household size and income impact food waste. Additionally, an assessment of behaviours can be analysed such as around food storage and meal planning. At the end of the questionnaire, informed consent was sought; all households taking part in the waste compositional analysis provided informed consent for their waste to be collected and analysed. Any information collected about the householder was treated as strictly confidential. The householder's details were not linked to the waste collected and it is not possible to identify the household within any published data.

The fieldwork was undertaken by Resource Futures (WCA) and Censuswide (questionnaire) November 2021 and March 2022. The two anonymised datasets (from the questionnaire and WCA) were linked using a unique identification number for each household.

The following section describes the detail of the methodology for the waste compositional analysis.

3.2 Sample design

The sampling process aimed at achieving a total sample size of 1,800 households, distributed within 12 local authorities. The sample was designed to be representative of UK households with regard to several factors. To achieve this, a three-stage sampling design was created: firstly local

authorities were selected, then collection rounds within these local authorities, and, finally, households within those collection rounds. The following sections describe the process, including the selection criteria, for each stage of this process in order to yield a sample that was representative of households across the UK.

Selecting local authorities to participate

The fieldwork was undertaken in England and Wales, reflecting the funding for the project. However, the sample was selected to be representative of the whole UK (for both the local authorities selected and the households within them). To make optimal use of the budget, the households in the sample were drawn from 12 local authorities, three in Wales and nine in England. This balances the need to have a dispersed sample and the increased logistical costs were the sample to be fully dispersed across the UK.

For the local authorities, a number of criteria were used to guide the selection:

- Include authorities with a mix of waste collection schemes – especially residual waste and any collections targeting food waste – that was representative of those in operation across the UK, as shown in Table 1 below;
- Include a range of deprivation levels and population density that is broadly representative of the UK; and
- Include authorities in England and Wales.

Groups of local authorities were formed using this information, from which local authorities could be randomly selected for inclusion in the detailed waste composition study. Specifically, local authorities were stratified by UK nation (England or Wales), waste collection scheme (i.e. frequency of food waste and residual collections), density and deprivation. Population density was described by the number of people per hectare for each local authority. Local authorities were then grouped into three quantiles. Similarly, deprivation was calculated by appending the average score for the Index of Multiple Deprivation (IMD) for each local authority; deprivation was also grouped into three quantiles.

The groups of local authorities containing the majority of LAs were then used to create the sample of LAs representative of the UK for the WCA (Table 2). Some groups of LAs were not included because they contained very few LAs therefore their inclusion would result in a sample which is not representative of the LAs in the UK.

WRAP contacted the randomly selected local authorities within each group, ensuring that there was agreement to participate in each group. Resource Futures followed up to obtain their permission to conduct the survey and waste composition analysis in their area.

Table 2: Local authorities included in the Waste Composition Analysis including information use to group each authority.

Country	Local Authority	Waste service type	Deprivation level	Population density
England	Bromsgrove District Council	No separate food waste, fortnightly residual	Low	Medium
	Wiltshire Council	No separate food waste, fortnightly residual	Low	Low
	Bedford Council	No separate food waste, fortnightly residual	Medium	Medium
	West Lindsey District Council	No separate food waste, fortnightly residual	Medium	Low
	Southampton City Council	No separate food waste, fortnightly residual	High	High
	Newcastle upon Tyne City Council	No separate food waste, fortnightly residual	High	High
	Richmond upon Thames London Borough Council	Weekly separate food waste, weekly residual	Low	High
	North Somerset Council	Weekly separate food, fortnightly residual	Medium	Medium
	Cheshire East Council	Weekly food (mixed organic), fortnightly residual	Low	Low
Wales	Monmouthshire County Council	Weekly separate food waste, fortnightly residual	Medium	Low
	Newport City Council	Weekly separate food waste, fortnightly residual	High	Medium
	Caerphilly County Borough Council	Weekly separate food waste, fortnightly residual	High	Medium

Table 3 shows how the sample of LAs included in the WCA closely matches to collection scheme types of the UK as a whole. For example, the proportion of local authorities in the UK operating a weekly food, fortnightly (or less) refuse collection service is 30%; the sample contained 4 local authorities with this collection system, which equates to 33%.

Table 3: Distribution of waste collection schemes for the sample and the UK population

Food waste collection:	Weekly		Fortnightly (or less)		No food collection
Residual waste collection:	Weekly	Fortnightly (or less)	Weekly	Fortnightly (or less)	Any
Sample (no.)	1	4	0	1	6
Sample (%)	8%	33%	0%	8%	50%
UK population	8%	30%	2%	12%	48%

Selecting collection routes and properties

Once the local authorities had agreed to participate in the survey, the project team gathered information about the waste collection routes, working with the LA to identify the most suitable rounds for the research spread across 5 working days and ensuring that these are within the same collection week. The number of houses, time and days of collection and where possible the actual route of the waste collection round was also identified.

From this information, Resource Futures determined which routes, within each of the 12 local authorities, would be best to include in the study to ensure that the households included would be representative of the UK. The ultimate intention was to make the waste routes selected within a waste collection type as representative as possible (within the confines of the study) of the UK by rurality and demographic classification (Output Area Classification[12]), and thereby minimise the use of post-weighting on the dataset.

In previous studies just five routes per local authority were selected to participate in the research. However, by selecting households across more routes it was possible to gather data from a wider range of areas at no additional cost. This helped ensure a more representative sample, reducing the need for weighting, and reducing uncertainty in the final results. This was achieved using postcode information about each of the waste routes obtained from local authorities and

[12] Output Area Classification (OAC) is the Office of National Statistics' (ONS) demographic classification system. It is based on the UK Census data and is an open source geospatial classification. OAC uses a combination of over 60 Census variables including average age, living arrangements, property type, ethnicity, immigration status, proficiency in English, presence of children and students, tenure, level of qualifications, vehicle availability, travel to work arrangements, employment status and sector of employment.

appending data existing data showing the OAC group of each postcode to every collection round. Routes were selected with the aim of being representative of each local authority included, whilst ensuring that the combined data of all twelve authorities would be as close to the UK profile as possible. Resource Futures identified around 3,000 households per local authority across the collection routes provided for interviews to ensure that response rates on the doorstep can be accommodated (including any reluctance due to the ongoing COVID-19 pandemic).

Flats or other properties with shared waste collection were excluded in the research. For this research, it is necessary to link information about food waste to individual (yet anonymous) households to help establish the link between demographic information and waste. This means that households for which waste cannot be identified (in a shared bin or waste receptacle) were omitted from the research. This is consistent with previous research.

Omission of flats was partially counteracted by weighting of the calculations (see Chapter 6.0), which takes into account that flats generally have fewer occupants than houses. However, this weighting does not take into account any other differences between flats and houses that may impact food and drink waste generated. The potential impact of this aspect of the methodology was assessed when this research was conducted in 2012 and found to have minimal impact on the results[13].

3.3 Questionnaire survey fieldwork

A face-to-face doorstep survey approach was adopted for the fieldwork. Previous research (*The Food We Waste* & *Household Food and Drink Waste in the UK 2012*) adopted an in-home, face-to-face approach. A face-to-face approach helped to ensure high rates of response and consent. In particular, a face to-face approach encourages consent to the waste composition analysis phase and allows the recruitment to this stage of research to be formalised.

The questionnaire took no longer than 25 minutes to conduct. A £5 shopping voucher was gifted to residents taking part in the survey to thank them for their participation. Survey fieldwork was conducted during the day and in the evening, along with weekend and weekday shifts to ensure that a cross-section of respondents participate in the research. Only those who are either solely or mainly responsible for food shopping or food preparation were included; those participants which were not responsible have not been included in the study.

In total, 3,380 interviews were carried out across twelve local authority areas. Of these, 2,919 consented to have their waste analysed. It was not possible to evenly distribute the interviews across all 12 authorities taking part due to the constraints of time, season and ongoing COVID-19 pandemic restrictions and associated staff shortages. The number of interviews conducted aimed to exceed the required number of households required for the waste composition analysis (1800

[13] Methods used for Household Food and Drink Waste in the UK 2012, WRAP, 2013, see chapter 12. Accessed via: Household Food and Drink Waste in the United Kingdom 2012 | WRAP

households). This ensured that there were enough households in each local authority for a sample which was representative of the UK. Additionally having extra households to sample from enabled effective route planning when collecting waste to be analysed, meaning households could be sampled on similar routes making daily collection of the required number of samples possible.

The fieldwork was undertaken in two waves, the first between 25th October and 17th December 2021, and the second between 4th January and 27th February 2022. This avoided responses to the questions being affected by the Christmas period. The effects of seasonality on food waste levels are discussed and quantified in the section Seasonality .

Questionnaire

The questionnaire design drew in the most part from questions asked on WRAP's regular household food waste questionnaire[14]. Questionnaire for participants of the Waste Composition Analysis details the questions used in this study. The questionnaire was completed electronically using tablets and with the aid of paper show cards for response options. Clear instructions were provided to the interviewer to ensure the data quality was as high as possible. The questionnaire was also translated into Welsh and respondents in Wales were offered the opportunity to participate in Welsh. In other ethnic communities where the target respondents did not speak English, the interview was conducted through another member of the household who did speak English.

Questionnaire data was collected by Censuswide on tablets using an online survey platform. The questionnaires were checked by Censuswide following completion by their surveyors, with additional checks being undertaken by Resource Futures.

Obtaining informed consent

Once householders had completed the survey, the consent form for the waste analysis work was introduced to them, having previously been mentioned in the introduction to the survey. If they agreed to participate, they were asked to sign a consent form[15], giving permission to the project team to collect and analyse their waste. Informed consent means that respondents a) know what participation in research entails, its purpose, how the results will be used, and by whom, and b) agree to participating in the research. This was explained at the start of the interview by the interviewer, outlined in the consent form, as well as in a supporting leaflet. The consent form and information leaflet were also translated into Welsh. A letter with the local authority's logo which highlighted their support for the study was used.

[14] UK Household Food Waste Tracking Survey 2022: Behaviours, attitudes, and awareness during a cost-of-living crisis, WRAP, Accessed: August 2023, Via: UK Household Food Waste Tracking Survey 2022: Behaviours, attitudes, and awareness during a cost-of-living crisis | WRAP

[15] Due to the ongoing COVID19 pandemic, residents were asked to sign the form electronically on a tablet rather than signing a paper form.

The performance of each interviewer was monitored, and best practice shared between interviewers. Additional support was provided to any interviewers who were achieving a lower-than-expected consent rate.

When the respondents agreed to take part in the study, their address and signature were collected on the tablet as evidence that they were willing to participate. A paper copy of the consent letter was left with the householder along with the information leaflet, which gave information about the project generally including FAQs and details of how data would be used. Importantly, the leaflet also contained a contact name, a direct free-phone telephone number and an e-mail address which recipients could use if they wished to contact someone about the survey or if they wished to withdraw from the study at any stage. The approach used resulted in only a very small number of those interviewed using the phone number or email address for further information or to withdraw their consent for the compositional analysis. Any who withdrew, did not have their waste collected for analysis and their survey data was deleted.

Care was taken in the phrasing of the leaflet to minimise any research effect on the respondent's waste behaviour. It is important to recognise that a survey of waste disposal behaviour could affect actual behaviour to some extent. In order to mitigate any research effect on the waste analysis work, a period of at least one weeks was left between the end of the survey period and the collection of the waste analysis data, and for most respondents this gap was longer, 2-3 weeks on average.

Confidentiality

Any information collected about the householder was treated as strictly confidential. The householder's details were not linked to the waste collected as unique codes for households were applied to the waste collected and the subsequent data, and it is not possible to identify the household within any published data. The householder's contact details were kept securely by Censuswide during the project and were deleted once the research was complete. Their address details were passed to Resource Futures in order to carry out the waste composition analysis, who also deleted them from their records once the study was complete. Where information has been passed to WRAP for the analysis in this report, it was anonymised, and it is not possible to identify individual households from this information.

3.4 Waste compositional analysis

This section details how the waste composition analysis was conducted by Resource Futures. The waste compositional analysis collected and analysed food waste from consenting households allowing the amount and types of this waste to be determined.

Collection of kerbside sample materials

For each household participating in the research, the waste streams collected and analysed were:

- kerbside residual waste; and
- any kerbside organic waste which is intended to include food (i.e., either separate food waste collections, or garden waste collections that accept food waste).

These two waste streams constitute the vast majority of household food waste collected by local authorities. A smaller proportion of food waste is also collected by local authorities as contamination in dry recycling and within HWRC (Household Waste and Recycling Centres) residual waste (see figure 2 in section 3.1 of the report: *Local authority collected household food waste in the UK 2022*[16]). Therefore, the composition of food waste collected by local authorities is well represented by sampling kerbside residual and kerbside streams targeting food waste.

Where collection of either residual or organic material was fortnightly, all relevant waste streams were collected and analysed over a fortnight period to take account of any variation in arisings between the two weeks. Where both waste streams were collected every week, the material from these households was collected during just one week.

The route and approximate timings of the collection were agreed with the local authorities in advance (see 'Selecting collection routes and properties' for details of the selection). Materials were collected from the target list of households derived from the survey stage of the research. Each day, a team of Resource Futures operatives visited the sample points and collected material in a non-compacting vehicle. The collection crew followed the sample round, as marked on their collection sheet, which were structured to ensure that the collection crew arrived well before the usual material collection, but not so early that households setting out their waste late were missed. A record was made of materials collected from each household, by waste stream, on the sample collection record sheet. Households were included if they had presented either residual waste or food waste in week 1. If in week 1 no waste was presented from a household originally targeted for collection another household was included instead. If in the second week of sampling a household did not present any waste, then no additional households were sampled. The material from each stream and each household was bagged separately, and marked with a tag to identify material stream, and with the household ID and number of bags (e.g. one of two) marked clearly on the tag, as well as on a label affixed to the bag as a backup.

Interaction with the public

The project team undertook a number of actions to ensure that the general public were not concerned during kerbside collections:

[16] Local authority collected household food waste in the UK 2022, Accessed via https://www.wrap.ngo/resources/report/local-authority-collected-household-food-waste-uk-2022

- Permission was gained from an appropriate member of each household during the survey phase (see 'Obtaining informed consent');
- All collection staff displayed photographic identification, and carried a letter of authority from the council explaining that they were authorised to carry out the work;
- The collection vehicles were equipped with magnetic signs with the logo of the relevant local authority;
- Uniform and personal protective equipment (PPE) were worn for safety reasons and so that the collection crew had a professional appearance;
- Local authority hotlines and the local police were briefed by the local authority to ensure that they were able to answer questions should residents contact them;
- An experienced site manager always accompanied the collection crew and talked to residents that may approach the team, explaining the purpose of their presence and giving them a letter of authority to keep (which included an appropriate contact name and number at the local authority);
- Resource Futures did not take waste that was not presented at the kerbside (i.e., no member of the collection crew would enter a garden to remove waste, except in one local authority where residents are instructed to present their waste within the boundary of their property, rather than at the kerbside); and
- The collection crew respected the wishes of a householder if they requested that their waste not be included. Reactive question and answers (Q&As) were drafted by WRAP and provided to the WRAP helpline at the beginning of the survey fieldwork period to maximise useful information provision to members of the public.

Sorting the waste

Each sort team was made up of a senior site manager, one supervisor (on all but the very smallest teams), and experienced operatives who sorted the waste, to make a total of up to ten people. Waste was sorted for each individual household and waste stream[17], to produce a set of results which linked to the household survey data. This enabled analysis of waste disposal behaviour by any variable in the survey (See: Questionnaire for participants of the Waste Composition Analysis), in addition to the combined sample result. Future work by WRAP will explore demographic and behavioural factors associated with household food waste.

Waste was weighed using digital scales calibrated by Bristol Weighing Machine on the 12th November 2021 & 17th February 2022 in accordance to UK Weighing Federation code of practice. These scales weighed waste to the nearest gramme. The process of sorting and data entry is summarised in Figure 1 and Figure 2 as well as being detailed below:

[17] The streams were collected, bagged and tagged separately from each household and then analysed. Therefore, each item of food waste was labelled with the waste stream from which it came (food waste, residual waste or food and garden waste).

- Each household was assigned a unique ID number when questionnaires were conducted, this number was tagged to each waste sample from that household allowing the waste for individual households to be recorded and associated with questionnaire data;
- The material from each household/waste stream was weighed and emptied onto a sort table. All items not comprising or including food waste were removed and weighed;
- Food items without packaging were bulked-up (i.e. placed in a single pile) by food category, sub-categories and food preparation state and weighed together (e.g. all fragments of potato peel together, all whole potatoes together); and
- Items or bulked-up categories were then weighed individually, and the relevant information entered into a tablet. The information is entered using a bespoke application created by Resource Futures with appropriate routing between questions to aid accurate data entry.

Where food items carried further information on the packaging, these were:

- Removed from the packaging;
- Packaging and food weighed separately;
- The food weight recorded as above; and
- All relevant and available details from the packaging recorded (including best before date etc.).

Using electronic data entry

Resource Futures used a bespoke application built using the software Fulcrum for data collection. As each item was entered, the information was uploaded to the cloud. This enabled quality checking of the data to take place concurrently with the data collection, and feedback given to the consultant on site when there were any concerns or anomalies with data. All weights were also entered on a paper form, so that the total weight of each sample was compared to the weight of the food in that sample (including packaging). Each item weighed constituted one item in the resulting dataset (Figure 8). Resource Futures undertook a number of pilot surveys to test that the entry form and results database were fit for purpose. After analysis, the separated material was disposed by the local authority.

Figure 1 Schematic of data collected relating to each item of food waste. This is part one of the diagram, data collected after food tertiary category is in the following schematic

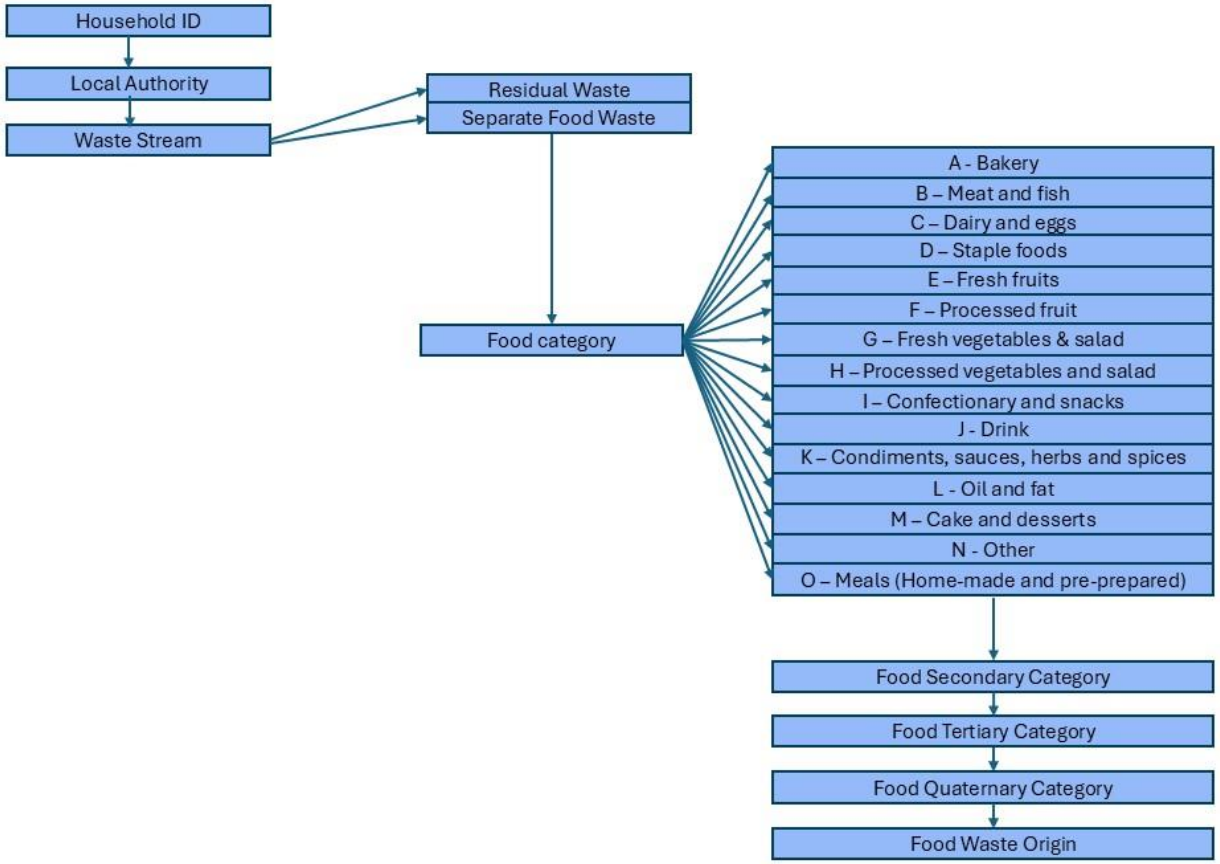
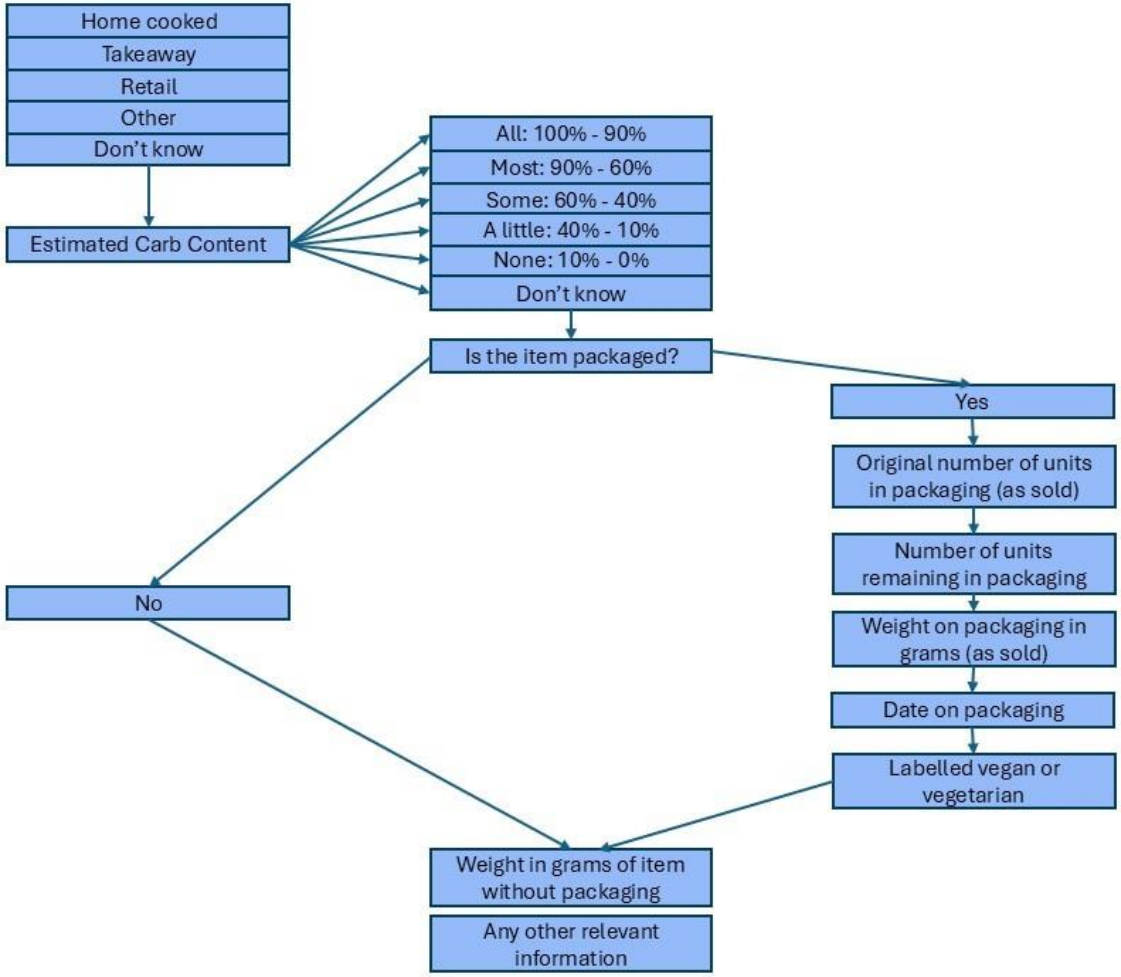


Figure 2 Schematic of data collected relating to each item of food waste after the food category data has been collected (see the previous schematic) This is part two of the diagram, showing additional data collected on each food item



4.0 Methods for Kitchen and Sewer Diary 22/23 research

This chapter of the report outlines in detail the methodology used to conduct food waste diary research. The diaries are used to inform the total amount and types of food wasted via the sewer and home composting, as well as the reasons for discarding for all household food waste (Table 1). As in the restated household food waste for 2007-2015 report[18], food fed to animals is no longer considered food waste. Therefore, although dairies record this data this is not a focus of this report and so is not discussed specifically in this section.

4.1 Diary research background

In 2007 WRAP first commissioned food waste dairies, with a single diary used to ask participants about food wasted via the sink, home composting and through local authority collected streams. In 2008, WRAP commissioned research focusing only on the amount of food and drink disposed to sewer – the report was published in 2009 under the name 'Down the Drain'. In 2012, with a view to update the evidence base on all household food and drink waste, a wider kitchen diary approach was utilised in the same way as the 2007 diary (i.e. asking respondents to record waste to destinations including the sewer, but also their residual bin, any separate food waste caddy and home composting etc.). WRAP concluded that this approach was not robust for estimating sewer waste as the resulting estimate of food and drink waste going down the sewer was around 75% lower than the 2008 estimate. Further investigation indicated that the increased burden on respondents (from being asked to record food and drink waste for all disposal routes from the home) was the most likely explanation of these discrepancies (see section 2.3 in the Methods used for Household Food and Drink Waste in the UK 2012 report[19]). It was therefore concluded that in future research, separate diaries should be used to capture waste down the sewer and waste in the bin/food caddy/compost.

At the start of 2022, with the objective of reviewing and updating the methods used to investigate food and drink waste to sewer, and the evidence base on household food and drink waste, WRAP commissioned a new study. A small pilot was undertaken with WRAP colleagues who tested an updated methodology which included a digital-diary prototype and the tried-and-tested pen and paper diary previously used.

[18] Household food waste: restated data for 2007-2015, WRAP, 2018, Accessed in August 2023, Via: Technical report templates (wrap.org.uk)
[19] Methods used for Household Food and Drink Waste in the UK 2012, WRAP, 2013, Accessed via: Household Food and Drink Waste in the United Kingdom 2012 | WRAP

In mid-2022, WRAP commissioned a third-party market research company, DJS Research, to conduct a full-scale study, taking into account observations and learnings from the previous 2008, 2012 and 2022 studies. Therefore, the agreed approach involved separate diaries for sewer waste and general kitchen waste. The sewer waste diary focused only food and drink wasted via routes such as the sink, macerators and the dishwasher. While the general food waste diary asked participants to record food disposed of via the:

- General waste bin,
- separate targeted food waste collections (mixed food and garden collections and separate food waste collections),
- home composting,
- feeding unwanted food to pets (note that since 2015 WRAP has not classed this as food waste)

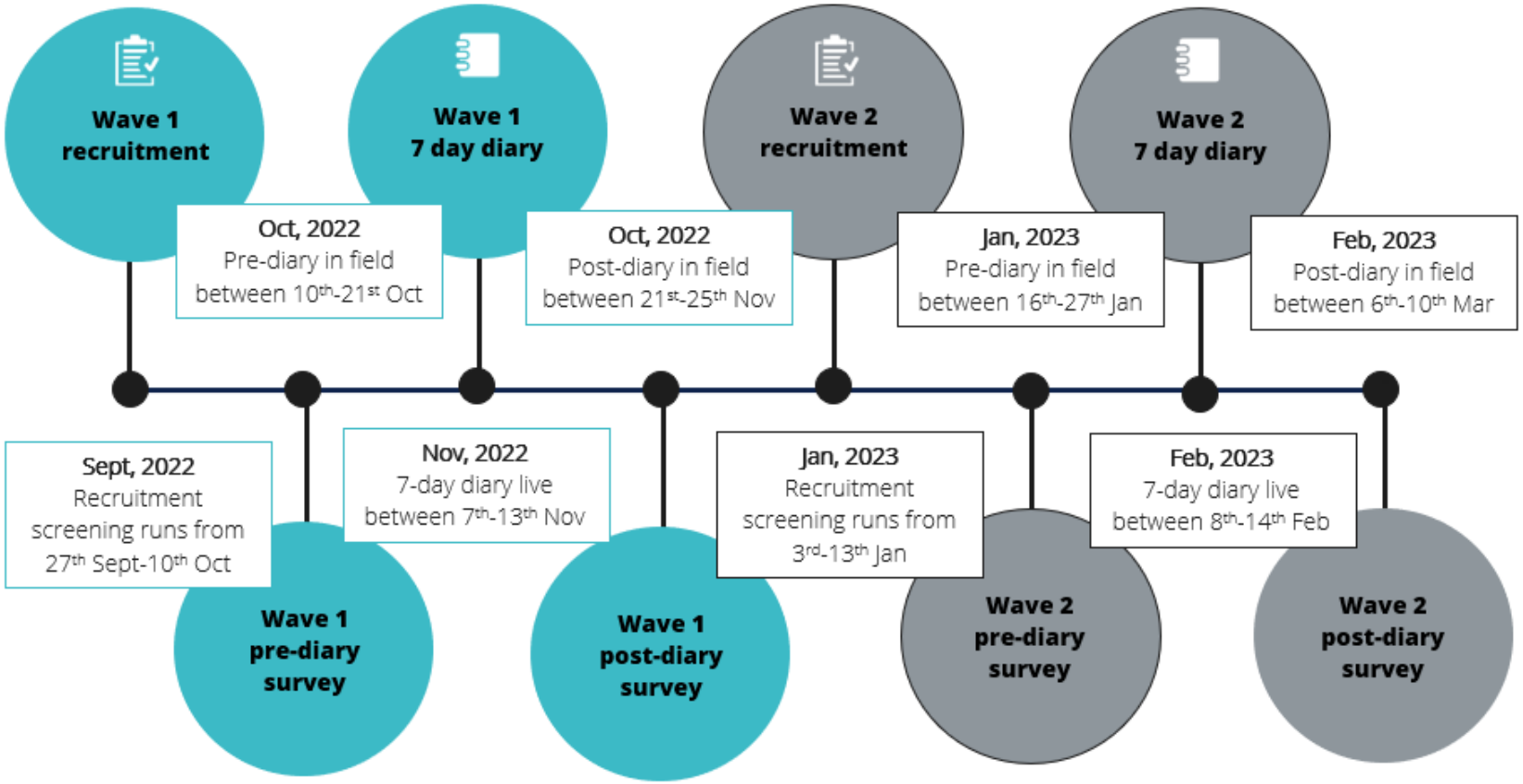
Two separate waves were carried out to record food and drink waste at different points in the year, including during the school holidays. Ideally studies looking at food waste would sample throughout the year to capture varying food waste behaviours as recommended by the Food Waste Index[20]. This study includes a wave in autumn (November 2022) and winter (February 2023) (Figure 3). Ideally further waves of research at different points in the year would have been conducted, but due to time constraints this was not possible. An adjustment has been applied to the data to account for some seasonal variation as detailed in section 6.5.

Although distinctively separate, the overall research methodology detailed below was replicated for each of the sewer waste and general food waste diaries. Revisions were made to the 2022/23 study's methodological approach based on the learnings from previous research carried out by WRAP. Subsequently, following learnings from wave one, further slight revisions were made to the research process in wave two.

The following section details the methodological approach for the 2022/23 sewer and kitchen diaries projects and the rationale for the approach taken.

[20] UNEP Food Waste Index Report 2021, United Nations Environment Programme, 2021, Accessed in: September 2023, Accessed via: UNEP Food Waste Index Report 2021 | UNEP - UN Environment Programme

Figure 3 Timeline and stages of the food waste diaries for both wave 1 and wave 2. Both the general kitchen diary and the sewer diary took place concurrently with different sets of participants.



4.2 Data collection methods

Recruitment/screening questionnaire

Before proceeding with recruitment, a screener questionnaire was prepared. Invites were sent via an online panel to a selection of individuals, inviting them to complete further screening questions, whereupon quotas were monitored to select or screen out respondents.

This questionnaire included essential screening criteria such as responsibility for food shopping and preparation, and availability during the scheduled 'diary week'. Profiling questions were also included, looking at gender, age, location, ethnicity, living situation, details about other household members, social grade, and access and use of food and drink waste collection services. These profiling questions were included to ensure a good fit with nationally representative quotas.

Additionally, the screener also asked respondents to indicate their preference for completing either a pen and paper or digital diary. In earlier pilots, prompts appeared to those who chose the latter to understand their digital confidence. However, as the sample was drawn from online panels, there were no anticipated issues with levels of digital competency and as such these prompts were removed.

The questionnaire also asked what, if any, appropriate measuring devices they own to ensure arrangements could be made to send either a digital set of scales or a 1 litre capacity measuring jug and 100ml capacity beaker before fieldwork began. For project budgeting purposes, a cap was set of no more than 50% of each sample (food and sewer waste) to require equipment.

Based on previous diaries research and a WRAP digital diary pilot project[21], a number of other revisions were also made based on the following rationale:

- Firstly, a discussion was had to determine whether to keep the recruitment screeners separate for the food and sewer waste projects or have one screener that covers both. The latter approach uncovered an opportunity to reach out to respondents who may not qualify for the food waste survey yet may still be eligible for the sewer waste, therefore speeding up the recruitment process. However, combining the two screeners would risk not being as explicit in asking respondents what would be expected of them. It was therefore decided to run both screeners independently of one another, but to align them in terms of the questions being asked, but with any food or sewer specific questions only being shown to relevant target respondents.
- As this work was carried out by DJS, a market research company on behalf of WRAP, It was discussed whether adding reference to WRAP's involvement in the research would

[21] This is an unpublished piece of research looking at the feasibility of running a digital diary research project. The recommendations of this work are included in the bullet points shown in this section of the report.

influence responses and potentially encourage respondents to modify their behaviour and reduce their food and sewer waste during the diary phase. However, ultimately it was agreed that including WRAP's involvement in the study added validity and trust in the project and was therefore included at the beginning of the screener.

- A pop-up diary entry illustration was added to the screener to show the types of entries respondents were expected to complete to ensure they understood the requirement and fully committed to the project from the outset to reduce the need for any top-up recruitment.
- In order to reflect the proportion of the population with access to food-waste collections (and usage of this service) as closely as possible, further relevant questions were added to the screener questionnaire.
- To ensure compliance with GDPR (General Data Protection Regulation) additional questions were added to obtain consent to share PII (Personally Identifiable Information) with third-party companies for distribution of paper copies or measuring devices, agreement with DJS research's privacy policies, permission for WRAP to share aggregated data in the public domain and for anonymised verbatims to be included in reports.
- The research needed to provide a full picture of waste for the whole household, but discussions flagged that some respondents may live in households where waste is managed separately by different individuals in the home (e.g. families with older children, house shares, etc.). For those completing the digital diary, functionality enabled each household to be provided with a unique ID and additional respondent IDs (if required) to each individual to record their own waste. A question was added to those who opted for digital diary completion to check whether the respondent intended to complete on behalf of the full household, or whether they required additional IDs for other members of the household. In the case of the latter, additional emails were requested and unique IDs sent to each email address. The pen-and-paper diary was more straightforward, as one booklet was sent per household which could be accessed by anyone in the household. However, additional information was included to emphasise the importance of capturing all food or sewer waste (as appropriate) and encouraging the respondent to inform others in the household to contribute too.
- To create further awareness of the commitment required, a question was added which advised respondents of the fieldwork dates to ensure they were at home for the full duration of diary completion. This was especially important as the second wave took place over a half term period to explore whether this had any impact on a household's level of food and/or sewer waste.

- To meet funding requirements of the Welsh government, a question was added to ensure all respondents living in Wales were asked for their language preference for the surveys, diary and other communications/notifications.
- Questions to collect personally identifiable information (PII) were added to the end of the screener for those who potentially need measurement devices sending out, to collect postcodes for analysis purposes and to have an email address for recontact once respondents had signed up.

Pre and post diary surveys

Before and after their diary completion, respondents were asked to complete a short (approximately 10 minute) online survey sent via an emailed link.

The aim of the pre and post diary surveys was to capture insight into shopping behaviours and habits within the household, as well as attitudes relating to food consumption and waste.

Prior to fieldwork, a review of the pre and post survey questionnaires used in previous WRAP diaries research was undertaken. It was decided that consistency with previous research projects must be maintained as direct comparisons between the data sets may be required. Therefore, any edits made to either the pre, and post diaries were minor, in line with best market research practise.

The following slight revisions were made to the pre-diary survey based on the following rationale:

- To enable a greater level of analysis from the data, WRAP's segmentation questions were included in the pre-diary questionnaire (as opposed to the recruitment screener as previously) to keep the latter short, concise and only inclusive of the essential screening questions.
- A question previously included in the recruitment screener measuring the percentage of food/drink waste in the home was moved to the pre-diary questionnaire to reduce the number of questions for the reasons mentioned above.
- In line with best practise, randomisation of answer codes was applied to questions asking about 'use-by' and 'sell-by' dates and the storage of food items to ensure randomisation in the order. This eliminates order bias and ensures that respondents aren't influenced by the order in which answers are presented in any way.
- During the questionnaire review, some inconsistencies in the questions asked between the food and sewer waste surveys were noted, and the order they appeared in. Edits were therefore made to ensure all relevant questions were included in both questionnaires and in the same order, to ensure greater consistency across pre/post data comparisons.

The following revisions were made to the post-diary survey based on the following rationale:

- During the second wave of the project, a question was added to the end of the post-diary survey asking whether respondents would be happy for WRAP to recontact them for the purpose of follow-up research between February 2023 and March 2024. GDPR considerations were taken into account and discussed with both WRAP and the third-party panel providers used by DJS Research during fieldwork.

Pen & paper diary

Prior to the 2022/23 study launching, a review of the pen-and-paper diary took place and the following amendments to the previous pen and paper diary were made:

- At an overall level, the general design (e.g. colours, fonts, imagery) were enhanced by the DJS Research creative services department.
- Textboxes were made bigger so that respondents had more space to enter details.
- The box which indicates reasons for no recording of waste was moved to the top of the page to make it more immediate.
- The box which allowed respondents to record the amount spent on food shopping on that particular day was amended to include a tick box that asked whether any shopping had been done, and what kind (main, top up, or impulse). This allowed for any future analysis looking at the correlation between waste and shopping instances.
- A calendar icon with the date highlighted inside was added to make it clearer which page should be completed on each day.
- Text "Need more space? Please continue any more entries you have on the next page." was added to make it clearer that there are two pages for each diary.
- A previous open ended question ("What? Description of the item and whether fresh, frozen, tinned, takeaway, homemade, ingredients used, etc.") was split out into a coded ("What food/drink type was this?) and open-end question ("What? Give a full description of what is thrown away, focusing on the main components/ingredients used.") to be able to capture whether the meal or beverage was homemade or store bought, and what ingredients went into the preparation. This amend made it clearer that this information was to be reported.
- Specifically relating to the sewer waste diary, WRAP informed the DJS Research team that some respondents had struggled with whether to include certain types of water waste in the previous study, and specifically brine/oil from some tinned foods. Clarification was therefore added to the instruction guide to confirm what should and shouldn't be included.

Digital diary

The aforementioned changes were also incorporated into the digital diary to ensure this closely followed the same structure of the pen and paper diary.

The pen-and-paper diary was replicated into a digital format whereby different forms/pages were included for the instructions, example entries, and dated pages for each diary day. The digital diary contained identical questions to the pen and paper using open text boxes, drop down menus, and multiple-choice answers to facilitate ease of completion online. Moreover, in order to mimic the pen and paper experience as closely as possible, a change was made to the diary platform in wave two which allowed respondents to have access to their previous diary entries as many times as they wished to make amendments as needed.

The only real difference was that the digital diary included functionality which allowed respondents to upload photos. This was encouraged if they had difficulty measuring the amount of food or drink they were disposing of and contributed to improving the accuracy of diary entries.

4.3 Sampling framework

A hybrid sample framework approach was undertaken which involved inviting a random selection of respondents from an online research panel from high-level targetable strata such as UK nations, gender and age. Research panels are a practical way to recruit participants who are generally willing to participant in research. Although it should be noted that research panels may include people who exhibit different types of behaviours than the general population. The alternative of recruiting participants from random households in the UK would be impractical because of the time needed to onboard new participants and ensure they were representative of the UK. Additionally, there is no evidence to our knowledge which suggests people in research panels produce significantly different amounts or types of food waste. Therefore, research panels are a practical and appropriate way to recruit participants.

Furthermore criteria which are known to influence the amount of food waste produced in a household were used in the sampling design, to ensure the sample is representative. Monitoring quotas were then set on other criteria, e.g. household composition, housing type and tenure, employment status and ethnic origin. Due to the constraints of the sample size, it was not feasible to interlock quotas in a more nuanced way (e.g. invite respondents who are White British, C1, Live in Birmingham in a mortgaged terraced house with four people in the household). However, a representative sample was still achieved by monitoring quotas and ensuring a good spread across the sample as a whole. Over-recruitment was carried out to ensure minimum numbers

were achieved. Sample sizes were driven by funding input from the devolved nations rather than being designed to be proportional to population size. However, final data tables were weighted to allow for accurate comparisons between nations (see section 6.0 for more information on weighting). Separate groups of participants were recruited for each of the diaries (sewer and general kitchen diary) and again different participants were used in each of the waves.

Therefore, the following sample sizes were proposed:

Sewer diary sample

To achieve 420 completes (pre-diary, diary entry, post-diary) over two waves across England, Wales and Scotland:

	Wave 1			Wave 1 TOTALS	Wave 2			Wave 2 TOTALS	TOTALS
	Eng	Scot	Wales	All nations	Eng	Scot	Wales	All nations	All nations
To achieve:	75	35	100	210	75	35	100	210	420
Recruit:	90	42	120	252	90	42	120	252	504

Food diary sample

To achieve 600 completes (pre-diary, diary entry, post-diary) over two waves, across England, Wales and Scotland:

	Wave 1			Wave 1 TOTALS	Wave 2			Wave 2 TOTALS	TOTALS
	Eng	Scot	Wales	All nations	Eng	Scot	Wales	All nations	All nations
To achieve:	135	65	100	300	135	65	100	300	600
Recruit:	163	78	120	361	162	78	120	360	721

Although, quotas were monitored during recruitment to ensure good fit with nationally represented figures for age, gender, region, household composition, housing type and tenure, employment status and ethnic origin, it was unlikely that this approach would exactly match the

national data in each aspect. As such, a weighting approach was designed and recommended (see section 6.3 for more details).

4.4 Participant engagement

Instructions

Right from the point of recruitment, instructions were included to make it explicitly clear what respondents were signing up for and the time commitment and timescales involved. When respondents then received either their pen and paper or digital diaries, the first section of the diary included clear instructions on what is required, hints and tips, dos and don'ts and a guide on how to measure the waste being thrown away. An example diary was also included to illustrate what was expected of respondents.

Help desk

For those who had any further queries, email helpdesks were set up for both kitchen and sewer waste projects to ensure respondents had a way to get in touch if needed. They were reminded of this in the aforementioned instruction pages. The helpdesk was contactable daily from 09.00 – 17.00, so respondents would get a quick reply to their queries either via telephone or email, depending on the nature of their enquiry.

Reminders

To remind and encourage participation, daily email reminders were sent to respondents. The content was amended depending on their engagement levels, for example, thanks were sent to those who had engaged with the diary and submitted entries, while a check-in was included for those who were yet to do so.

Further to this, daily texts were sent including fun food related facts to remind respondents via a different medium. Understanding that respondents have varying commitments, both texts and emails were sent at differing times of the day to increase the likelihood of them being seen at least once by respondents.

Incentives

As in previous studies, an incentive was offered to respondents who completed all aspects of fieldwork (pre-diary, diary, and post-diary). This was increased to £60 in GiftPay vouchers, compared to £50 offered previously in 2008 to try and maximise engagement. Incentives are extremely effective in encouraging participation and are often used as part of research projects. In this case, it was made clear to respondents that in order to receive the incentive, all three stages of fieldwork needed to be completed in full.

4.5 Logistics

Measuring devices

The recruitment screener gathered data on the number of respondents who required measuring devices (either digital scales or measuring jug/beaker) along with their address details and permission to share with third-party companies. These were then sourced, packaged and sent to the relevant respondents via third-party pack and send company. During wave one of the research, postal strikes resulted in the need to use couriers rather than the Royal Mail.

Paper and pen diaries

Similarly, paper versions of the diary were sent well in advance of fieldwork beginning. Diaries were designed using the InDesign programme before being securely shared with a third-party print and send company, with consent being given by respondents during the recruitment screener.

Welsh speaking respondents

For those completing the diary in the Welsh language, a third-party translation company was used for the digital diary as there were no Welsh speaking respondents who opted for the pen and paper approach.

4.6 Fieldwork

Challenges

As with any research project, challenges arose during wave one which were then reviewed prior to wave two. In the majority of instances, appropriate changes were made prior to wave two (see the 'adaptations along the way and rationale' section for more detail), however other broader observations are useful to note for any future study.

With additional time needed for the pen and paper diaries (i.e. printing, packing and sending), those completing fieldwork via this approach received their diaries and therefore instructions before those who opted for the digital diary. This meant that some respondents therefore had longer to read and process the instructions, allowing them more time to consider any questions or concerns they may have in advance of completing the diary. For those completing the digital diaries, questions had to be raised on the very first day of diary completion (when the diary went live), which may have discouraged participation from some respondents. The impact of this is hard to assess as most participants opted for the digital diary (958 vs 122 via the paper diary), the

small sample of paper diaries used therefore makes it hard to make meaningful comparison between the diary types (this is explored in section 'Digital diaries').

The email helpdesk proved a helpful way of logging issues that respondents faced during fieldwork. During wave one, most of the emails related to what food or drink should or shouldn't be included, along with how to record this in the diary. Conversely, the main query raised in wave two was respondents having some initial technical difficulties accessing their online diary which was quickly resolved by the helpdesk.

Adaptations along the way and rationale

A review of the first wave of fieldwork resulted in the following adjustments prior to the second wave:

- During the first wave of the research, a high proportion of drop-out was experienced between recruitment and the pre-diary survey exercise. To mitigate the recurrence of this risk in wave two (particularly as initial timing would have resulted in the Christmas/New Year break between recruitment and the pre-diary survey), new timescales were designed to reduce time lag between each stage of the project. Grouping the recruitment screener and pre-diary survey was considered as one way of ensuring higher completion rates from recruitment screener through to the pre-diary survey. On balance, this was decided against as keeping the two surveys separate allowed the research team to address/replace any respondent drop-out at the pre-diary survey stage, reducing the risk of respondents simply completing a combined first stage, but dropping out at diary completion (when it would be too late to replace them).
- A number of queries were received from respondents between the pre-diary survey and diary stages of the project who wondered what the next steps were as this was considered a large gap from their perspective. It was therefore agreed that timescales were to be updated to create a shorter gap between stages during wave two.
- In addition to this, queries were received throughout wave one from respondents asking when they would receive their incentive payment. During wave two, additional emails were sent to provide more detailed updates and reminders about all three stages to ensure respondents were kept informed. Each time they completed a step, an automated email was sent, reminding them of what would happen next. Further to this, visuals were added to all communications to show the full fieldwork process, where respondents were up to, and what was still needed for them to reach their incentive.
- A number of respondents in wave one wanted to change or check their diary entries retrospectively, either because of an error, or to ensure it was submitted properly. Functionality was added during wave two to allow respondents to see all previous entries and make any amends if required.

- During wave one, respondents who had opted to complete the pen and paper diary were also provided with access to the digital version. As a result, a number changed their approach prior or midway through fieldwork which resulted in a lower than anticipated number of pen and paper completions. Therefore, no access to the digital diary was provided to those who opted to complete the pen and paper version in wave two and this risk eliminated. Instructions were updated to reflect this, specifically within the version for those completing via pen and paper who were encouraged to contact the helpdesk if they wished to change.
- The "finish study" button at the end of the digital diary during wave one was updated to "submit diary" during wave two. Respondents understood the former to mean that they would finish all remaining stages and would no longer receive their incentive so were reluctant to press the button resulting in many diaries entries appearing incomplete during data cleaning (when this wasn't the case).
- At the end of the first wave, the full set of data was cleaned and analysed where it was revealed that a number of respondents had incorrectly inputted some entries. The main error being respondents inputting all waste (food and sewer) when they had been assigned to only input one waste route. It was therefore decided that daily checks were to be carried out during wave two so that respondents could be contacted quickly to correct their future entries.

4.7 Learnings and recommendations

In order to avoid pen and paper respondents receiving their diaries before the digital diary respondents (as stated in section 4.6, efforts should be made to coordinate email delivery of the digital diary and postage of pen and paper diaries. This way, all respondents receive their diaries at the same time and have plenty of opportunity to read the instructions and forward any initial queries to the appropriate helpdesk.

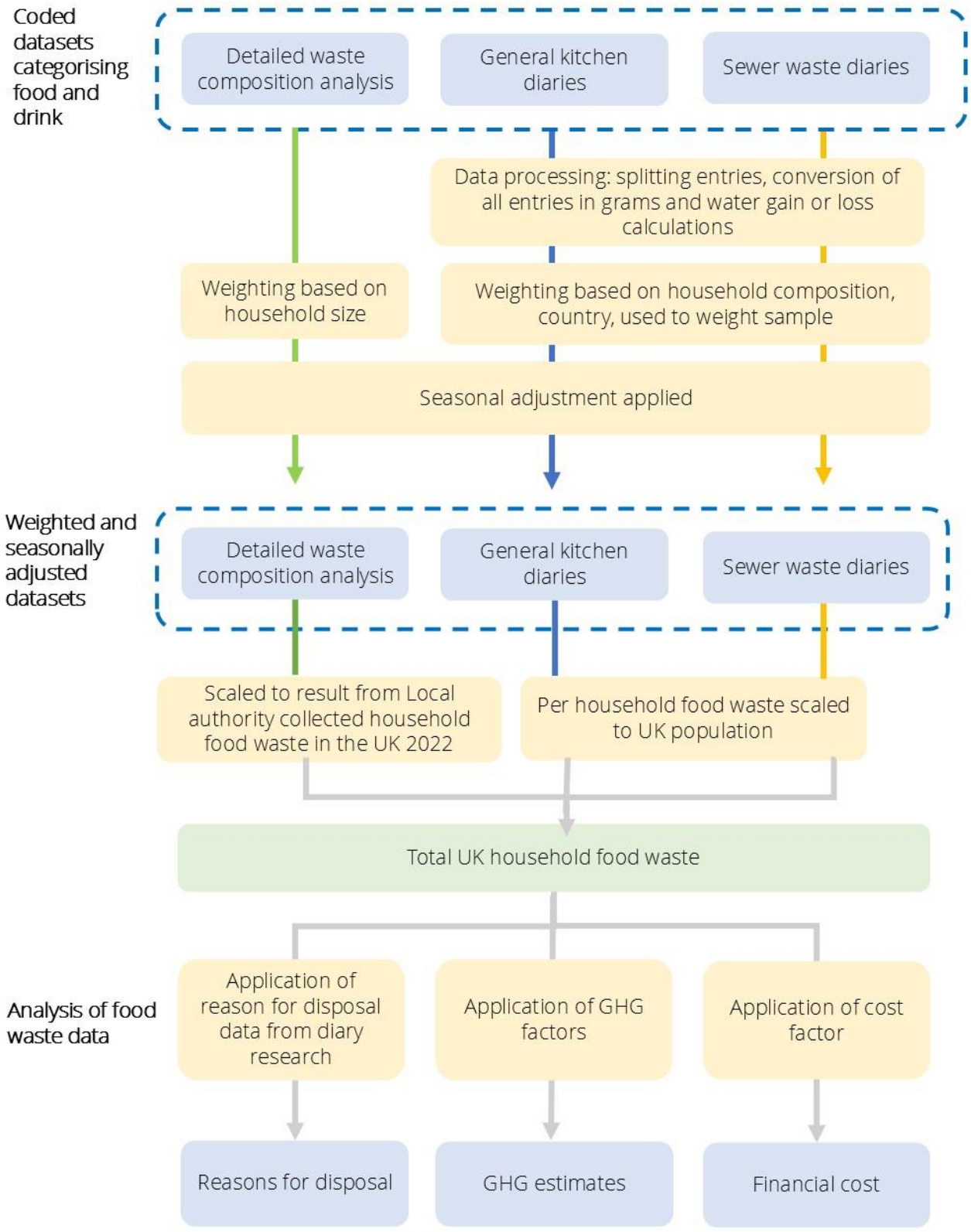
A number of emails were sent to the helpdesk querying what items can and cannot be included in both the kitchen and sink diaries (as stated in section 4.6). While this is clearly stated in the instruction pages this was obviously overlooked by some respondents, a short video showing an example of a respondent carrying out the tasks and filling in the diary might make this clearer to respondents.

Further to this, the dos and don'ts section of the instructions should be reviewed to ensure they are being understood correctly. Carrying out a cognitive exercise with potential future respondents would be beneficial to ensure they are clear and fully understood prior to launch.

5.0 Overview of calculations

This chapter details how the calculations were performed to obtain estimates for Household Food and Drink Waste in the United Kingdom 2022. This section only outlines how the total amount of food waste is calculated, and the types of food wasted. This data is in turn used in a number of further pieces of analysis outlined in Chapter 8.0. A flow chart detailing the calculations performed on each dataset to arrive at a total UK food waste estimate and the subsequent analysis is shown in Figure 4.

Figure 4: Flow diagram outlining datasets used and calculations performed.



5.1 Calculations for local authority collected waste

For the local authority collected waste, the detailed information by food type comes from the detailed waste compositional analysis. This includes detailed information on what food is wasted and in what proportion, for example, it provides detail on the proportion of waste which are vegetables and salads, as well the proportion, which is potato, whether that is whole, mashed or just peel. This detail can then be combined with total amount of food waste estimated by the Local authority collected household food waste in the UK 2022 analysis.

The waste composition dataset was weighted by the number of occupants in the household as described in section 6.2. The weights of food were also seasonally adjusted using the method outlined in 6.5. Seasonal factors were calculated to convert information collected in one month of the year to an annual figure, as different food items are bought and wasted in different quantities through the year.

The total amount of food and drink waste was then scaled so the total amount is equal to the estimate from the Local authority collected household food waste in the UK 2022. This scaling was necessary as the latter research provides the more robust estimate of the total amount of food and drink waste in local authority collected waste streams (see section 2.2 for more details). The scaling factor used was 1.24, i.e. estimates in the detailed compositional analysis needed increasing by 24%. This is very similar to the scaling factor used for 2012 estimates in Household Food and Drink Waste in the UK which used a factor of 1.23.

Households that were interviewed and consented to have their waste collected but did not have any waste collected and analysed were excluded from the sample. This decision was made because the reasons for the waste not being collected were varied. For instance, some households did not present waste at kerbside for reasons that are unknown. For others, it was not possible to link the waste to a household; this happened when bins and bags were placed or piled together making identification impossible. Had the reasons all been connected with non-presentation of waste (inferring a waste prevention effect), then there would have been advantages to leaving the households in the sample, but recording their food and drink waste as zero. However, as this wasn't always the case, they were excluded.

5.2 Calculations for diaries research: home composting and sewer waste

For food and drink waste that is home composted or disposed of via the sewer, data from the General Kitchen Diary 21/22 and Sewer Diary 21/22 research was used. As described in 6.3, weighting factors relating to multiple factors were applied and the results scaled to the total number of households in the UK. These results were then seasonally adjusted using the method

in section 6.5. The following section describes details of the processing of the kitchen-diary data prior to weighting and seasonal adjustment.

Diary data processing

Before analysis of the diary data could be undertaken a number of data processing steps were required, as detailed in the following sections.

Splitting entries

Firstly, entries were split which contained more than one food item. Participants of the diaries were asked to only enter one food item per entry into the diary and to record what was thrown away, the quantity discarded and the reasons for discarding. However, a number of entries contained information for multiple different food categories. These entries required splitting out to allow a record of the weight of each individual food category to be recorded. In cases that entries were split it was assumed that the reasons for discarding and disposal route was the same for each of the items included in one entry.

Weight of items

Where items were recorded in the diary as the number of items or as a volume, these entries required converting to a standard weight.

In cases in which participants recorded the number of items wasted, for example the peel of an orange or whole broccoli. Standard weights were determined for such items by using data readily from supermarket websites which provide pack weight and serving suggestion information. Data was not always available for every item and so proxy values are used in some cases in which similar items are used to represent items with missing weight data. The weights used to for different food items are shown in section 10.2 in the appendix.

To make diary entries as practical as possible for participants, they could record the quantity of food or drink wasted using a range of measures, including handfuls, measuring jugs and spoons. In these cases the volume of each type of measuring device was used to calculate what was wasted in mls. The bulk density of specific types of food items was then used to determine the weight of these entries. Bulk density data was collected from a range of sources including

scientific papers[22][23], data originating from the FOA[24] and data published for use by food processes[25].

Calculations relating to water gain and loss

When certain food is cooked and drink products are made they can gain and loose water. For example, pasta and rice absorb water, while meats loose water, water is then added to drinks like tea. This a consideration to make when using data relating to the quantity of a food category when it is purchased, such as in the calculations outlined in the sections: Percentage of food purchased wasted analysis and Food and drink price.

For products which are cooked, firstly the percentage of the wasted category which is cooked is calculated using data from the waste composition analysis which recorded this information for every item recorded in the analysis. For this cooked waste, a change in weight is applied from factors given in McCance and Widdowson. This gives the equivalent amount of raw food for these categories, which is then added to the weight of food thrown away as raw to understand the purchase weight of food wasted.

For products in which water is added, such as for hot drinks (tea and coffee), gravy and stock, the amount of water added was estimated using serving suggestion information provided by brands and supermarkets. This enabled a proportion of each of these liquids to be calculated as being the dry purchase weight of that food/drink category.

5.3 Calculations to determine edibility

Household food waste statistics were restated for 2007 to 2015 in which food waste was re-classified as being either edible or inedible, rather than using the previous classification based on avoidability. The approach is outlined in detail in the report 'Household food waste: restated data for 2007-2015'[26], this study uses the same assignment of edibility as used previously. The approach is summarised in this section.

The terminology used is with the edibility classification system, which is as follows:

[22] Jahanbakhshi, A., Abbaspour-Gilandeh, Y., Ghamari, B., & Heidarbeigi, K. (2019). Assessment of physical, mechanical, and hydrodynamic properties in reducing postharvest losses of cantaloupe (Cucumis melo var. Cantaloupensis). Journal of Food Process Engineering, 42(5), e13091.
[23] Pa, N. F. C., Chin, N. L., Yusof, Y. A., & Aziz, N. A. (2014). Power ultrasound assisted mixing effects on bread physical properties. Agriculture and Agricultural Science Procedia, 2, 60-66.
[24] [kg/m³] kg-m3.com All about density, Accessed in June- July 2023, Via: Material volumetric mass density (kg-m3.com)
[25] Bulk Density, Machine and Process design, Accessed in June- July 2023, Via: Bulk Density Values | Food Processing Reference (mpd-inc.com)
[26] Household food waste: restated data for 2007-2015, WRAP, 2018, Accessed in August 2023, Via: Technical report templates (wrap.org.uk)

- Food waste: Food and the inedible parts of food removed from the food supply chain (or household) to be recovered or disposed of (including - composted, anaerobic digestion, incineration, disposal to sewer or landfill). This definition excludes waste prevention activities, namely redistribution for human consumption, or diverted to feed animals[27].
- Wasted food (also referred to as edible parts): products (or parts of products) intended for human consumption. This includes material which is still suitable for consumption when it is disposed of (i.e. would be regarded as 'edible') and that which may no longer be suitable for consumption at the point of discarding (e.g. for example due to it passing a 'use by' date or being spoiled). It excludes inedible parts.
- Inedible parts: Components associated with a food that are not intended for human consumption. Examples of inedible parts associated with food could include bones, rinds, and pits/stones. "Inedible parts" do not include packaging. What is considered inedible varies among users (e.g., chicken feet are consumed in some food supply chains but not others), changes over time, and is influenced by a range of variables including culture, socio-economic factors, availability, price, technological advances, international trade, and geography.

Therefore, these three terms are related according to the equation below:

$$Food\ waste = Wasted\ food\ (edible\ parts) + Inedible\ part$$

When this approach was being designed there were two main considerations to make when classifying food categories according to edibility:

1. How to determine whether a food category is considered edible or not. There was not an existing framework or set of guidelines outlining the edibility of food. Therefore, a questionnaire was used to understand the culture norms in the UK around what food categories are considered edible or not.
2. A number of food categories are partially edible and partially inedible. For example, a banana skin is inedible, but its flesh is edible, so when a whole banana is discarded a proportion is edible and a proportion is inedible. Therefore, a particular food category needs to be assigned an edibility percentage, to determine what proportion of that product is edible.

[27] The FUSIONS project, through extensive consultation, published a food waste 'definitional framework' and definition which is largely consistent with this definition, in terms of covering both food and inedible parts and the relevant destinations. See the following: FUSIONS Definitional Framework for Food Waste 2014.pdf (eu-fusions.org)

Determining the edibility of a food item

To determine if a food category is considered edible or not a questionnaire was used to classify food items which are edible and inedible. Two questions were developed, the first asks which items / parts of items people actually eat, and the second about which items they consider edible and which inedible, whether or not they eat them themselves. The two questions were asked about sixteen different parts of items (e.g. apple peel, parsnip peel). These items were chosen because a) they represented a substantial amount of waste as previously determined from waste compositional analysis and diary research, b) their 'edibility' was judged to be 'borderline' by the authors, and c) they could be used as proxies for other items.

A score was assigned to the answers given to each question based on the response (e.g. 1 for "always", 1/3 for "occasionally") and average taken across the respondents' answers for each item to obtain a score for each item for each question. For a given item, the average score was calculated for the two questions, providing a single value to reflect its perceived edibility from the UK population.

Of the items in the questionnaire – the following items were classified as edible (i.e.

food):

- Crusts of bread slice
- End slices of loaf
- Apple skin
- Cooked chicken skin
- Potato skin
- Bacon rind
- Broccoli stalk
- Cauliflower stalk
- Outer cabbage leaves
- Carrot skin

The following were classified as inedible parts:

- Apple core
- Orange peel (including the zest)
- Cabbage - stem & hard centre
- Parsnip skin

- Oil drained from a fish tin

For items not asked about in the survey, a similar item was selected as a proxy where possible. For example, parsnip peel was considered similar enough to swede peel (i.e. the peel of a root vegetable, often cooked in a similar way and with a similar type of peel) to be used as a proxy. Rules of thumb were developed to cover situations where no similar proxy was available or where it might not make sense to use one (Table 1). These rules were based on the purpose and use of the item. For example, used oil could have been from deep frying (where the purpose of the oil had been fulfilled once used for frying). This case was then compared with similar use cases in the survey. In the example of used oil, it was classified as like oil drain from fish tins, which had been classed as inedible parts by the survey participants representing the UK public. This led to used oil being classified as an inedible part.

To see the details of how all food types have been classified in this study and the 2015 restated household food waste study, see Appendix D in the Household food waste: restated data for 2007-2015 report[28].

Table 4: Rules of thumb for determining inedible parts

Rule	Examples
Only a small amount of weight (mass) is extracted by processing/cooking	*Boiling bones (for stock), brewing tea leaves, unchewed gum*
Used as a storage medium for its primary purpose	*Brine, pickling vinegar, oil in fish tins*
Fulfilled its primary function before discarding	*Cooking oil, chewed gum*
Seasonal produce prepared in a specific way according to many recipes	*Outer leaves and ends of Brussels sprouts*

[28] Household food waste: restated data for 2007-2015, WRAP, 2018, Accessed in September 2023, Via: Courtauld 2025 baseline and restated household food waste figures | WRAP

Breakdown of items containing edible and inedible parts

When considering an item of waste that contained both edible and inedible parts, a method was needed to determine approximately how much of it should be assigned to each category. A hierarchy of sources was used to determine this. Each contains a percentage for the edible weight of a wide range of common foods:

A. Food Standards Agency. (2002). McCance and Widdowson's The Composition of Foods, Sixth summary edition. Cambridge: Royal Society of Chemistry.

B. Lynch, F.T. (2011). The Book of Yields: Accuracy in Food Costing and Purchasing, Eighth edition. Hobeken, New Jersey: Wiley.

C. US Department of Agriculture, Agricultural Research Service, Nutrient Data Laboratory. (2015). USDA National Nutrient Database for Standard Reference. Release 28. Slightly revised May 2016. [online] available at: https://www.ars.usda.gov/northeast-area/beltsville-md/beltsville-humannutrition-research-center/nutrient-data-laboratory/docs/usda-national-nutrientdatabase-for-standard-reference/ [Accessed 11/12/2017]

Edibility calculations relating to carcass meat and bones

The waste composition waste analysis provided details about the % of meat on bones and carcass meat. For instance, where a chicken carcass has been thrown away, the proportion of meat left on the carcass was assessed visually and recorded according to the bone carcass details in Table 5. An edibility percentage could subsequently be assigned depending on the proportion of meat on carcass. The edibility percentage assigned are shown in Table 5 which is the midpoint of the range given in the bone carcass details.

This data could therefore be used to determine the edibility of individual items in the detailed waste composition analysis. The average edibility of different meat categories could then be calculated from the waste composition data and was applied to the meat categories in the diaries data set.

Table 5: Bone carcass details assigned to meat products in the Detailed Waste Composition analysis and the edibility percentage assigned to products depending on the bone carcass details given.

Bone carcass details	Edibility % assigned
Majority meat (60%+)	80%
Around half meat (40-60%)	50%
Some meat visible (20-40%)	30%
A little meat visible (10-20%)	15%
Very little meat visible (1-10%)	5%
No meat visible (<1%)	0.5%

5.4 Total UK household data

Total household numbers in the UK are used to help calculate total food waste in the UK (to scale up per household figures from the diaries) and to calculate average per hh food waste arisings.

Household estimates for 2022 are from the Office for National Statistics Families and households dataset.[29].

In previous years projected and modelled numbers were utilised. Table 6 shows the data sources used in previous studies and in this study. Total household numbers have a low impact on total food waste estimates as they are only used for understanding food waste from home composted and sewer waste streams, which combined make up 15% of all household food waste.

[29] Families and households - Office for National Statistics

Table 6: Household estimate sources used in current and previous studies

Estimate	England	Wales	Northern Ireland	Scotland
Household Food and Drink Waste in the UK 2012	DCLG household projections	Growth in number of households for each nation between 2011 and 2012 (from DCLG projections) applied to 2011 census data		
Synthesis of Household Food Waste Compositional Data 2018	ONS 2016-based household projections used to scale up 2015 estimates			
Household Food and Drink Waste in the UK 2022	ONS Families and households dataset			

6.0 Weighting and seasonal adjustment

6.1 Objective of weighting in this study

Weighting of the sample data is used to adjust where the sample does not match the population for important variables, in this case variables that correlate with the amount or type of waste being thrown away. The use of weighting factors helps to eliminate bias in the results but also reduces the effective sample size and leads to worse precision. Weighting changes the relative contribution of certain values in a sample, meaning some samples are slightly increased and others decreased depending on whether they are over or underrepresented. The effective sample size then measures the degree of precision, showing what sample size could have been used to provide the sample level of precision. An effective sample size will always be lower than the actual sample size, although weighting should be used to create a representative sample whilst having as small an impact on the effective sample size as possible.

Different elements of this research have different requirements for weighting because they are sampling at different scales:

- Local authority collected household food waste in the UK 2022: as this work synthesises together information from individual local authorities, the key is to ensure that the local authorities in the study are as representative of the UK population of local authorities as possible (or the geographical area they are representing).
- Waste compositional analysis and diary research: in these studies, the total amount of waste generated by each participating household is measured or estimated. Therefore, ensuring the households in the study are representative of the UK population of households is the key factor.

6.2 Weighting of Waste Composition Analysis

The detailed waste composition data has been weighted to household size, which is consistent with the weighting applied to the analysis in 2012[30]. The number of people in the household has been demonstrated to correlate strongly with the amount of waste produced in many studies (e.g. Household Food and Drink Waste in the UK 2009 and 2012). Within the waste compositional analysis, there is an underrepresentation of one-person households which is likely driven by flats not being included in the waste composition analysis. Flats often use communal containers,

[30] WRAP, Methods used for Household Food and Drink Waste in the UK 2012, 2013, Accessed via: Household Food and Drink Waste in the United Kingdom 2012 | WRAP

making it difficult to understand one household's waste. Other sized households are either marginally over or under-represented this likely a result of chance and because one-person households are under-represented, meaning other household sizes become over-represented (Table 7). Weighting factors have therefore been applied to the dataset according to the size of each household.

Table 7: Percentage of different household sizes in the sample data and in the UK. This data is used to create the household size weighting factors.

Household Size	Sample data	UK actual	Weighting factor
1 person	20.3%	29.5%	1.45
2 persons	43.3%	34.1%	0.79
3 persons	16.3%	15.8%	0.97
4 persons	13.2%	14.2%	1.08
5+ persons	6.7%	6.3%	0.94

6.3 Weighting of kitchen and sewer waste diaries

In this study the diaries research has been weighted to household size only (Table 1). Household size has been shown to influence the amount of food wasted by a household and has been used in previous WRAP research to weight household food waste data[31]. In the diaries, one person households were underrepresented while all other household sizes were slightly overrepresented. While household composition was monitored when the sample was designed, this underrepresentation of one person households still occurred. The weightings applied resulted in an effective sample size of 595 for the general kitchen diary compared to 620 as the actual sample size. For the sewer diary the effective sample size was 429 compared to 459 as the actual sample size. Estimated sample size provides an indication of the precision of the survey sample, showing what sample size could have been used to provide the sample level of precision. Therefore, an effective sample size which is only slightly lower than the actual sample as shown here indicates a more precise sample.

[31] WRAP, Methods used for Household Food and Drink Waste in the UK 2012, 2013, Accessed via: Household Food and Drink Waste in the United Kingdom 2012 | WRAP

Table 8: The percentage of households in the sample by household size compared to the UK as well as the weighting factor applied to data from each household size

Diary	Household size	% of sample	% in the UK	Weighting factor
General kitchen	1	21.6%	29.5%	1.37
	2	36.7%	34.1%	0.93
	3	16.1%	15.8%	0.98
	4	16.7%	14.2%	0.85
	5+	8.9%	6.3%	0.72
Sink	1	19.4%	29.5%	1.52
	2	36.8%	34.1%	0.93
	3	18.7%	15.8%	0.84
	4	16.3%	14.2%	0.87
	5+	8.7%	6.3%	0.73

Another factor considered for weighting the diary data was country, as the sample over represents households from Wales relative to the proportion of Welsh households in the UK. Sampling was designed in this way to reflect the funding for this project. In previous household food waste work, the country of the UK has not been used to weight the sample as this is not typically a factor which is a large influence on food waste arisings. Additionally, applying a weighting factor based on both country and household size on the effective sample size is large. Specifically, the effective sample size would be reduced by 46% for the sink diary data and by 58% for the general kitchen diary. This would mean a less precise sample and an overreliance on a smaller set of households.

6.4 Stratification of local authority collected household food waste data

In the 2022 Local authority collected household food waste in the UK project, the residual food waste estimates were calculated by stratifying the local authorities according to whether they offer a separate food waste collection or not. Local authorities offering a targeted food waste collection tend to have a lower proportion of food waste in their residual stream compared to authorities which don't offer a separate food waste collection. Stratification in this way is

consistent with the methodology applied in previous years. Stratification means that residual waste composition data from local authorities with separate food waste collections was only applied to tonnages from local authorities with a separate collection.

6.5 Seasonality adjustment

The amounts and types of food purchased are known to vary seasonally. This is shown in monthly data from 2007 to 2020 provided by the Family Food survey. It is likely that more waste is generated for a given type of food when more of that food is entering the home. Therefore, purchasing data was used the adjust the amount of the amount of specific waste food categories. The same methodology was applied as in 2012 increasing comparability between the studies. The 2012 methodology provides more detail on the impacts of seasonality and how these impacts were investigated[32].

Both the Waste composition analysis and diaries research requires a seasonal adjustment. The waste composition analysis was conducted over a 5-month period (November 2021 and March 2022). The diaries in this study took place over a 4-month period (November 2022 and February 2023). Therefore, both of these pieces of research provide a 'snapshot' in time of the waste produced by households for a particular point in time has therefore been adjusted to ensure the data is representative of the whole year. The Local authority collected household food waste in the UK analysis does not require a seasonal adjustment as it utilises waste composition data collected throughout the whole year and Waste Data Flow tonnages which show annual waste arisings.

To adjust food categories according to seasonal variation in food purchases and so waste, Family food data was utilised between the years 2007 and 2020. A time series analysis was undertaken, with a second-order polynomial trend-line fitted to purchasing data of a particular item. For each month, the deviation from this trend-line is calculated as a percentage of the trend-line value for the month in question. These deviations are averaged for each month giving the seasonal trend in purchases. These trends can in turn be utilised to determine factors that convert estimates of waste for one month to an annual estimate. For example, if purchasing levels were 10% lower in the month of the year that research was undertaken compared to the annual level of purchasing, then the estimate from that month would be divided by 0.9 (i.e. 100% – 10%) to obtain the average for the year.

These seasonal adjustments are undertaken at different WRAP category levels. This study uses different levels of categories which vary in their detail, with the less detailed categories having subcategories. For example, a category 1 is Fresh Fruit, a category 2 Banana and a category 3 in

[32] Methods used for Household Food and Drink Waste in the UK 2012, WRAP, 2013, Accessed via: Household Food and Drink Waste in the United Kingdom 2012 | WRAP

Banana Peel. WRAP categories were assigned to DEFRA categories mainly at the category 2 level, with some assigned at the category 3 level. In some cases, the WRAP and DEFRA categories did not align, with the DEFRA category encompassing more than one WRAP category, in these cases new categories were created to be used for mapping to DEFRA categories. For example, a WRAP category was created including onions, spring onions and leeks. In total 8 new categories were created, which encompass multiple category 2s to allow mapping to the DEFRA family food categories. Most items are seasonally adjusted at the category 2 level, 75 of the 95 category 2s are used for seasonal adjustment. For Beef and Pork these categories were adjusted at the category 3 level. In total only 6 categories were not adjusted to season, generally these were more general categories such as 'N.3 - Mixed food from multiple categories' and 'H.3 - Other root vegetables'.

7.0 Diary underestimation

This chapter outlines why diaries are likely to be under estimating food waste which is home composted and disposed of via the sewer. The chapter also assesses how the diary should subsequently be used and reported.

7.1 Why is there underestimation in the diaries?

Diaries are one of the few options for collecting data on household food waste disposed of via the sink and via home composting. Diaries are, however, suspected of underestimating food waste compared to waste composition (WCA) studies. A study by WRAP (Quested et al. 2020[33]) suggested that the levels of underestimating are not stable and vary depending on context and the design of the diary.

WCA is a more accurate way of measuring waste as researchers can directly measure what is being thrown away rather than relying on participants to record their waste, as with diaries. However, these methods are problematic for home composting and sewer waste. For example, a composition analysis of a compost heap would have issues of knowing when something was discarded and as food decomposes it would be hard to identify.

Recent data collected in the detailed waste composition analysis (WCA) (2021/22) compared to the diaries (2022/23) supports this idea (

Table 9). When comparing the WCA data (collected in later 2021 and early 2022) directly to the diary data (later 2022 and early 2023), the result is a 42% underestimation, a little higher than the underestimation in 2007 (40%) and much higher than in 2012 (30%). However, the level of under-estimation for 2021-23 is actually closer to 34%, when you take into account the fact that Local Authority collected food waste fell by 8% between 2021 and 2022, when the majority of WCA was

[33] Quested, T. E., Palmer, G., Moreno, L. C., McDermott, C., & Schumacher, K. (2020). Comparing diaries and waste compositional analysis for measuring food waste in the home. Journal of Cleaner Production, 262, 121263.

collected, and 2022, when the majority of diary data was collected. This is level of underestimation (34%) is broadly in line with previous studies.

Table 9: LA-collected food waste per household: level of underestimation between data collected from diaries and WCA.

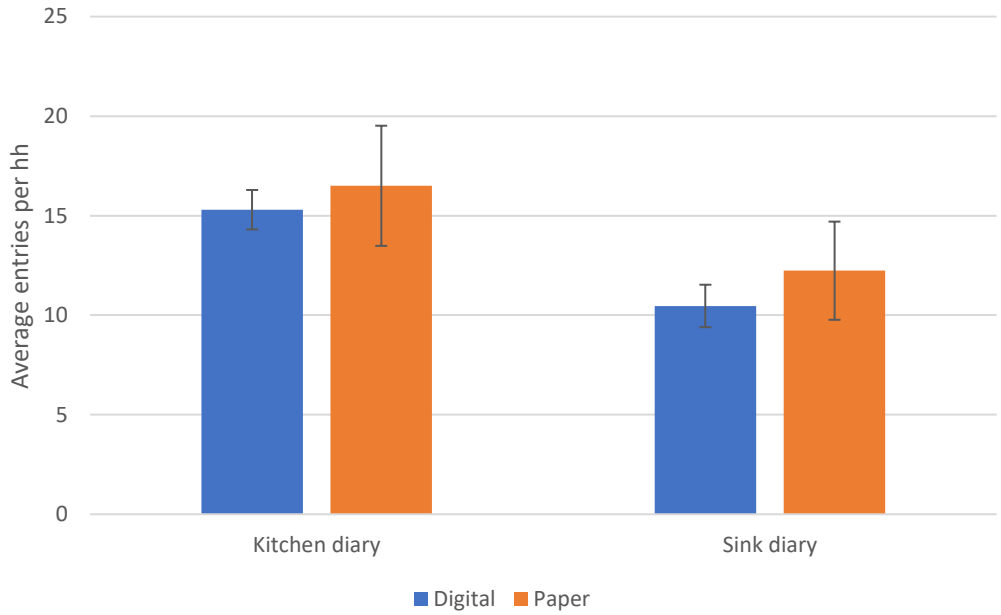
Amount of LA-collected food waste	Diary	WCA	Level of underestimation
	kg/hh/wk	kg/hh/wk	(%)
2021-23	1.45	2.52	42%
2012	1.89	2.69	30%
2007	2.18	3.63	40%

Digital diaries

One of the differences between this diary and previous diaries is that participants had the option to fill in this diary electronically. Most took this option, rather than use paper diaries. This section explores whether this change in mode had any impact on the results. Having the physical pen and paper diary in the kitchen where food and sewer waste occurs might have served as a tangible reminder to complete entries.

Figure 5 shows that, on average, the number of entries submitted via the pen and paper diary was higher than the digital diary for both waves across both food and sewer waste during the 2022/23 diary study. However, the confidence intervals, shown as error bars in suggest that this difference is not significant. It's important to note that far fewer respondents opted for the pen and paper diary (122 vs 958 via digital diary), which is why the confidence intervals around the paper averages are larger. Therefore, another study designed to compare the impact of using paper diaries compared to digital diaries is required to understand the effect of digital diaries further.

Figure 5: Average number of entries per household in the kitchen and sewer diaries using both digital and paper diaries. The error bars show the confidence intervals around each average.



Representation of composting households in the diaries sample

The representation of households which use home composting in the diaries data may impact the estimates made for the total food waste home composted.

As the amount of food home composted per household is scaled up to the total number of households in the UK, the method relies on the proportion of composting households in the diary to be the same as that of households which compost in the UK. Over or under-representation of composting households in the diaries would result in an over- or under-estimation of the total food waste home composted. Table 10 shows that compared to other WRAP surveys, the proportion of home composters in the 2012 and 2022/23 diary studies varies from the expected proportion.

Table 10: The percentage of households which compost in the diaries studies and according to the WRAP tracker surveys

Year	Source	% households composting
2012	WRAP Diaries study	29.5%
	WRAP tracker (unpublished)	23.1%
2022	2022/23 WRAP Diaries study	16.8%
	Recycling questionnaire (unpublished)	19%

Has sewer and home composted waste declined?

Taken at face value, the results indicate a decrease in amount of food waste disposed of via the sewer (48% compared to 2012) and home composting (68%). This section discusses other data to help us understand whether a downward trend has occurred. In summary, whilst small reductions could have occurred, any change is likely to be much smaller in magnitude than the diaries data suggests.

The apparent decreases would indicate a substantial behaviour change, which is not supported by other studies or datasets to our knowledge. For compost waste, the food waste tracker surveys suggest a reduction in the proportion of households home composting by 4% percentage points between c. 2012 and 2022 (a decrease of around 20%).

For sewer waste, WRAP is currently engaging with water companies to understand if they have any evidence for changes in food waste over time. It is understood that water companies have run campaigns to reduce food waste down the sink, which may have reduced food waste. Potentially, these could have led to a diversion of *some* food and drink waste (oils & fats, more 'solid' waste) from sewer disposal to LA collection. However, it is hard to get information on the prevalence of such campaigns over time or their impacts. It is unlikely that these campaigns would have influenced drink waste being poured down the sink or into the dishwasher, as most people would not pour those into the bin or separate food-waste collection, so even if they have changed behaviours in some households, it is unlikely to have reduced sewer waste on the scale seen.

Alternatively, there is the potential that people have reduced the types of waste going down the sewer or to home composting (an *at-source reduction*, rather than diversion to other waste streams). This would be unlikely as i) food waste disposed of to these destinations has broadly similar reasons to LA collected waste; ii) we would expect to see a corresponding reduction in sales of drinks and other relevant food categories: this has not been observed.

The increase in food waste collected by local authorities *could* also be used as supporting evidence for reducing sewer and compost waste. Food waste could have been diverted from the sewer and home composting to waste streams collected by local authorities. Food waste collected by LAs has increased by 0.3 million tonnes since 2012. However, this increase is more likely explained by a number of factors, such as the pandemic changing behaviours and low food prices prior to the cost-of-living crisis. The diaries suggest that in total sewer and compost waste has decreased by 1.1 million tonnes since 2012, three times the increase in food waste now collected by local authorities. Therefore, the reduction in sewer and compost waste cannot be explained by the diversion of food waste to local authority waste streams alone.

Overall, there may have been a change in food wasted home composted or going down the sewer, but any reduction is highly unlikely to have been of the extent suggested by taking the data at face value: this apparent change is not supported by other evidence available. This analysis

leads us to the conclusion that food-waste estimates derived from diary data are insufficiently accurate to track changes over time, and there LA collected waste should be used when tracking HHFW.

7.2 Options for measurement – what data to use for sewer and home composting

While we do not use diary data to track trends over time in this study, we still propose to use diary data to present a complete picture of the types and reasons for HHFW. A number of data adjustments were considered to account for known biases. However it was decided to use the diary data without a data adjustment as the difference between the diary data and waste composition data) in 2021-2023 was comparable to the underestimation for the 2007 and 2012 studies (see *Table 9* above). No adjustment factor was applied in these previous studies.

7.3 Options for reporting diary data

In addition to how to best use diary data, we considered three options for how to report data. The reporting options considered are shown below with the pros and cons of each outlined in Table 11.

A. For reporting trends, use only data relating to LA collected waste; for detailed information (e.g., for campaigns), also include sewer and home composting
B. Only report LA collected waste for both trends and detailed information
C. Include all disposal routes for reporting trends and detailed information

Option A was considered the best option as it allows the use of robust LA data for assessing trends over time and still enables the use of the collected diary data. Although the diary data is suspected of underestimating food waste, its use allows a more complete picture of food waste in the UK when looking into details around reasons for discarding and for understanding drinks waste which is mainly captured through diaries.

Table 11: Pros and cons for different options around how to report diary data

Option	Pros	Cons
A	Strong trend data – can infer changes over time from Local authority collected household food waste in the UK results Allows us to present complete picture of food and drink waste (types of waste, reasons for	Have 'two' food waste figures: one for trends (only including LA collected waste) and a broader one where we present the detail. Need to use the correct one in different circumstances – requires clear guidance internally and externally.

	discarding, costs, GHG emissions, etc.), including all relevant disposal routes.	Does not allow comparison of other waste streams (but we don't have robust data to do this anyway).
B	Have one number for HHFW in all contexts All data is relatively robust / strong	Doesn't give full picture: e.g., milk and other drinks will be greatly underestimated due to no sewer waste reporting Does not allow comparison of other waste streams (but we don't have robust data to do this anyway)
C	Have one number for HHFW in all contexts	Large reduction in total HHFW that we don't believe is substantiated by the evidence

8.0 Analysis of household food waste data

This chapter outlines different pieces of analysis utilising data showing the total amount of food waste in the UK as well as the types of food wasted.

8.1 Reasons for discarding analysis

The diaries data was used to determine the reasons for discarding. Each time a participant recorded what and how much of a particular item was discarded, the reason it was discarded was also recorded. This allowed a reason for discarding for each specific food item to be known. Table 12 shows all of the reasons participants had the option to specify and how they have been grouped into broader reasons used in the analysis.

Table 12: Reasons for discarding used in reporting and which of the original reasons given by participants in the diary responses have been grouped into the reported reasons for discarding.

Reason for discarding used in reporting	Original reason for discarding given by participants
Smelled/ looked off	Smelled/ looked off
Past date on label	Past date on label
Cooked, prepared or served too much	Cooked/ prepared too much
	Served/ poured too much
	Served too much
	Went cold
	Went flat/ cold
Accidents	Accidents (burnt etc.)
Personal preference/perceived inedible	Didn't fancy it/ rejected it
	Inedible/ no use for this
Residue from pan/bowl/packaging	Residue (e.g. bottom of the pan)
	Residue (e.g. bottom of pan)

For each food category the reasons for discarding in terms of a percentage split by weight were determined. These percentages were then applied to the total weight of food wasted by category level, to give the total amount of food wasted by reason for discarding.

8.2 Percentage of food purchased wasted analysis

To analyse the percentage of food purchased wasted, the amount of food purchased in the UK is divided by the amount of food wasted.

To understand the amount of food purchased in the UK the DEFRA Family food data was utilised[34]. At the time of analysis, the most recent data available was for 2022/23, which was therefore used in this analysis. The DEFRA family food data uses a different set of categories used by WRAP and so the two categorisation systems were mapped to one another to allow edibility percentages and wasted quantifies to be associated with purchased quantities. The amount of food purchased which is edible was calculated by using the same edibility percentages applied to food waste data (see section 5.3). For meat categories with a bone, different edibility percentages were applied to purchased food compared to wasted food as the amount of meat on bones is lower for wasted items. To understand the edibility of meat products with bones as they were purchased data McCance and Widdowson was utilised.

The quantity of food wasted was adjusted to represent the weight of each item as it was purchased, as in the process of consumption some food categories loose or gain water. This is outlined in more detail in the section Calculations relating to water gain and loss .

8.3 Food and drink price

8.3.1 Financial cost to households

To calculate the price to UK households of wasted food and drink data from the DEFRA Family food survey[35] was utilised as used in 2012 research. The DEFRA Family food survey contains data on weight and price of purchased food by UK households. This data is used to calculate the cost per edible gram of food purchased by dividing the sum of the purchase weight by the sum of the expenditure for each food category. To work out the cost of the edible portion of food, this cost per gram is divided by the edibility % so that the cost of a food item is only assigned to what is typically eaten.

To enable the use of the DEFRA family food survey, the food categories used in this study were assigned to the DEFRA food categories. This study uses different levels of categories which vary in

[34] 2022/23 Family food datasets, accessed via: Family food datasets - GOV.UK (www.gov.uk)
[35] 2022/23 Family food datasets, accessed via: Family food datasets - GOV.UK (www.gov.uk)

their detail, with the less detailed categories having subcategories. For example, a category 1 is Fresh Fruit, a category 2 Banana and a category 3 in Banana Peel. WRAP categories were assigned to DEFRA categories mainly at the category 2 level, with some assigned at the category 3 level. In some cases the WRAP and DEFRA categories did not align, with the DEFRA category encompassing more than one WRAP category, in these cases new categories were created to be used for mapping to DEFRA categories. For example, a category was created including onions, spring onions and leeks.

The quantity of food wasted was adjusted to represent the weight of each item as it was purchased, as in the process of consumption some food categories loose or gain water.

8.3.2 Financial cost of food waste treatment for local authorities

To quantify the financial cost of food waste treatment for local authorities the cost for treating a tonne of waste was applied to the estimated total amount of food waste collected.

The *WRAP Gate Fees report 2022-23*[36] was used for the cost of treatment per tonne of waste. Table 13 shows the figures used in the analysis as well as the assumed disposal method for each stream. For residual waste, 87% of residual material is sent to Energy From Waste (EfW) and 13% to landfill as calculated using data from Waste Data Flow (WDF). For targeted food waste collections, food can be collected completely separately from all other material in which case it assumed to be sent to Anaerobic Digestion (AD). Alternatively targeted food collections can be mixed food and garden collections, in which case they are assumed to be treated via In Vessel Composting (IVC).

For the total tonnes of food waste collected at the kerbside, data from the Local authority collected household food waste in the UK was utilised[37].

[36] WRAP Gate Fees report 2022-23, WRAP, 2023, Accessed via: https://www.wrap.ngo/resources/report/gate-fees-report-2022-23
[37] Local authority collected household food waste in the UK 2022, Accessed via https://www.wrap.ngo/resources/report/local-authority-collected-household-food-waste-uk-2022

Table 13: The cost per tonne of treating waste from specified waste streams via an assumed disposal method

Waste stream	Assumed disposal method	Cost per tonne (including transport)
Residual	Energy From Waste (EfW) (87%)	£122
Residual	Landfill (13%)	£156
Food only collection	Anaerobic Digestion (AD)	£31
Mixed food and garden collection	In Vessel Composting (IVC)	£79

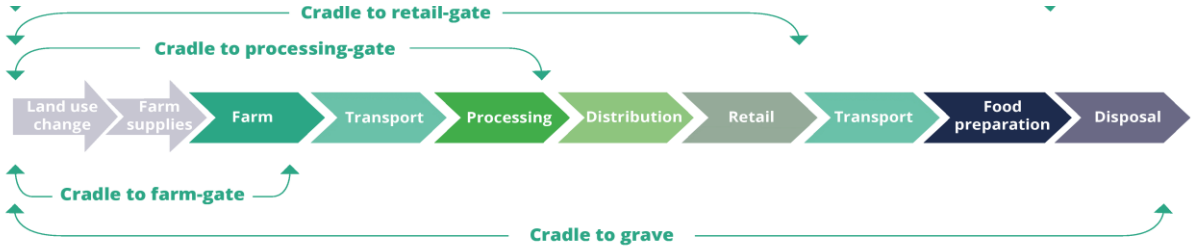
8.4 Calculations for greenhouse gas emissions

The greenhouse emissions associated with wasted food in the UK was assessed to help quantify the environmental impact of household food waste. The analysis aimed to look at the impact of food that has been wasted rather than an assessment of the GHG savings by reducing food waste as conducted, for example, by Philippidis at al. (2019)[38]. In this analysis, GHG emissions associated with the full supply chain from Cradle to Grave have been calculated for individual food categories. However, it should be noted that emissions from land use change are inconsistently included in the sources of data used in these estimates, therefore the estimate produced in this analysis are likely to be a slight underestimate. Land use is inconsistently included by studies and databases used for GHG estimates because of the complexity and uncertainty involved in quantifying emissions from land use change. The diagram shown in Figure 6 shows all the elements which have been accounted for in determining the emissions related to different food categories analysed in this study. Scope 3 GHG emissions data published by WRAP[39] was used to determine the emissions from cradle to processing gate. The emissions relating to food from

[38] Philippidis, G., Sartori, M., Ferrari, E. and M`barek, R., Waste not, want not: A bio-economic impact assessment of household food waste reductions in the EU, RESOURCES CONSERVATION AND RECYCLING, ISSN 0921-3449, 146, 2019, p. 514-522, JRC114250.
[39] Scope 3 GHG Measurement and Reporting Protocols for Food and Drink, WRAP, Accessed in: January 2023, Accessed via: Scope 3 GHG Measurement and Reporting Protocols for Food and Drink | WRAP

distribution to disposal have been calculated using data underpinning the GHG model used for Tracking UK Food System greenhouse gas emissions[40].

Figure 6: A diagram showing the lifecycle of a food product. Each stage in this diagram has emissions related to it which have been accounted for in the GHG analysis. Land-use change and is inconsistently included in the data sources used for the estimates used in this study.



To calculate the emissions data from cradle to processing gate the emissions related to specific food products were assigned to the food categories used in this study. The scope 3 GHG emission database[41] provides kg CO2e factors per kg of a range of food products. The scope 3 database compiles emission factors from a number of sources. A data quality framework is provided in the database. Using this, it was decided to remove references with a poor data quality score unless this was the only reference available for a particular food category. Some of the food categories used in household food waste study were not included in the database (e.g. meals, soup, coleslaw, other vegetables). Table A1: (in the Appendices) shows the categories that a proxy was needed for and the other similar food category used as a substitute factor.

To calculate emissions related from distribution to disposal the WRAP GHG model[42] was utilised alongside the DEFRA Family Food Survey[43]. The WRAP GHG model provides data on an average GHG factor for a typical tonne of food. For this work, some further detail was gained from looking at the emissions relating to products which were either refrigerated or cooked. The refrigerant emissions in the GHG model, as well as energy related to household chilling appliances (refrigerator, freezer) were applied only to 'refrigerated' foods, and the household energy use for cooking appliances then only being applied to the cooked foods. All other emission stages are applied evenly across all foodstuffs. This forms an intermediary detail level as it doesn't look at the

[40] Tracking UK Food System Greenhouse Gas Emissions: 2022 Update, WRAP, Accessed in: January 2023, Accessed via: Tracking UK Food System Greenhouse Gas Emissions: 2022 Update | WRAP
[41] Scope 3 GHG Measurement and Reporting Protocols for Food and Drink, WRAP, Accessed in: January 2023, Accessed via: Scope 3 GHG Measurement and Reporting Protocols for Food and Drink | WRAP
[42] Tracking UK Food System Greenhouse Gas Emissions: 2022 Update, WRAP, Accessed in: January 2023, Accessed via: Tracking UK Food System Greenhouse Gas Emissions: 2022 Update | WRAP
[43] 2020/21 Family food datasets, accessed via: Family food datasets - GOV.UK (www.gov.uk)

specific appliances used or specific cooking methods, but this level of detail is difficult to apply to the food waste data accurately.

Subsequently the GHG emissions for each individual food category was calculated from cradle to grave by combining cradle to processing gate and distribution to disposal GHG data. The factor created for each category was then multiplied by the edible weight of that category wasted by households to provide the GHG emissions for each food category. The GHG emission footprint has therefore been allocated wholly to the 'edible parts' of food waste on the logic that the edible parts are where the 'value' to consumers is.

8.4.1 Comparisons with previous household food waste GHG emissions

The analysis of GHG emissions related to household food waste was conducted with the intention of showing emissions for food waste in 2022. The analysis was not conducted with the intention of showing changes in emissions between years, however this section looks at why the estimate for 2022 is likely to vary from previous years. The reasons for differences in GHG estimate between years include both methodological differences and differences caused by changes in the GHG emissions of food waste.

Comparison with GHG emissions using 2012 household food waste data

The greenhouse gas emissions associated with wasted food and drink in 2012 were estimated to be 19 million tonnes of CO_2 equivalent which rises to 25 million tonnes of CO_2 equivalent when including the GHG from indirect land use change[44]. The greenhouse gas emissions (GHG) associated with wasted household food and drink in the UK accounted for approximately 16 million tonnes of CO_2 equivalent in 2022.

In 2012 a bottom-up approach was used to estimate the emissions from wasted food, in which GHG emission factors are assigned to each food group and scaled up according to the weight of wasted food from each group. A bottom-up approach was also used for the 2022 estimates of GHG emissions however there are some key differences in the calculations used for the estimates between years. The possible reasons for differences are explored below.

[44] Greenhouse gas emissions associated with wasted food were also estimated for 2015 in WRAP (2017) - *Household food waste: restated data for 2007-2015.* However, the 2015 calculations used a top-down calculation of greenhouse gas emissions per tonne of wasted food was used, and it has been observed in previous calculations that a top-down calculation results in a larger estimate than a bottom-up calculation. The 2012 figure used a bottom-up calculation for farm-to-processing-gate emissions and is therefore more comparable to the 2021 figure.

Edible waste tonnages

The difference in the amount of edible household food waste between years could play some part in the differences in GHG emissions between 2012 and 2022. In 2022, the estimate for edible household food waste is approximately 9% lower than in 2012.

Difference in composition of waste between years

Analysis has been conducted to understand whether these differences in GHG emissions are due to the types of food thrown away. The analysis suggested that this was not the explanation.

Lower emission factors

The emission factors used (tCO_2e per tonne of food waste) in 2022 are lower than those used in 2012 (Table 14). This subsection explores some of the key reasons for 2021/22 emission factors being lower than those used in 2012.

Table 14: GHG emission factors (tCO_2 per tonne) used in 2012 and 2022 for food categories associated with the highest emissions in 2022 from edible food waste.

Category	2012 tCO_2e per tonne	2022 tCO_2e per tonne	% difference in factor used
O.2 - Composite meal	9.1	6.2	-32%
B.2 - Beef	86.2	36.5	-58%
B.5 - Poultry (chicken / turkey / duck)	9.0	6.7	-26%
B.1 - Pork / ham / bacon	11.8	10.1	-14%
G.15 - Potato (all chips in processed)	2.4	1.7	-29%

Land use emission factors

The land use change assumptions used in 2012 estimates and 2022 estimates are a notable difference between the studies.

For 2022 land-use change (LUC) emissions refer to those that occur when the demand for a specific land use results in a change in carbon stocks on that land, due to either a conversion

from one land-use category to another or conversion within a land-use category[45]. The land-use categories include forest land, cropland, grassland, wetlands, settlements, and other lands. This means that 2022 GHG estimates look at the impact of direct land use change. However, the 2022 estimates use data from a range of sources, which inconsistently include LUC in GHG estimates due to the complexity and uncertainty linked to LUC estimates. For 2012 the estimates for land use change includes both direct emissions connected to crops consumed and indirect emissions arising from the effect of land use for consumed crops displacing other crops to agricultural land obtained by land use change. The previous estimates therefore include indirect land use change. The inclusion of additional emissions (i.e. indirect land use change) and consistently applying land-use change to all food categories, is likely to be a key reason for higher emission estimates compared to 2022.

Therefore, the methodological difference of how emissions from land use are quantified in 2012 and 2022 explain some of the differences in the factors used. Further work will be conducted to understand the land-use change emissions associated with household food waste in 2022.

Decarbonisation of the electricity grid

The decarbonisation of the UK electricity grid also partly explains the difference in emissions factors used in the 2012 and 2022 estimates. The lower carbon intensity of electricity in 2021/22 means that emissions related to parts of the food value chain which are significant electricity consumers will be lower compared to 2012. Food manufacture, retail and households are all associated with higher electricity consumption, and a decrease in the GHG intensity of energy has been a key driver of emission reductions across the UK food system since 2015[46]. This sub-section looks at the GHG emissions associated with elements of the food value chain with a large electricity consumption in 2012 and 2022.

To analyse whether a specific part of the value chain is associated with more GHG emissions in 2022 compared to 2012, then disaggregated emission data which shows the emissions related to a specific part of the value chain is required. Disaggregated emission data is available for most elements of the value chain used in the 2012 data, however disaggregated data is not available for all elements of the value chain in the 2022 data. Therefore, this section compares emissions related to the following parts of the value chain:

- Distribution to retail, retail and transport to home (with the emissions from each element combined into one factor),
- home refrigeration,
- cooking.

[45] Scope 3 GHG Measurement & Reporting Protocols, WRAP, 2022, Accessed via: Scope 3 GHG Measurement and Reporting Protocols for Food and Drink | WRAP

[46] UK Food System GHG Emissions, WRAP, 2021, Accessed via: UK Food System GHG Emissions | WRAP

Table 15 shows the emission factors used in the 2012 and 2022 studies to determine the GHG emissions of household food waste, as well as the percentage difference between the factors used for different elements of the food value chain. The emissions related to cooking shows a large decrease (53%) in 2021/22 compared to 2012. While emissions related to home refrigeration are 14% lower in 2022 and emissions related to distribution to retail, retail and transport to home combined only reduced by 3%. The lower reduction in emissions from distribution to retail, retail and transport to home maybe partly because emissions related to transport have been combined with emissions from retail. Transport emissions have been noted in the *UK Food System GHG* report to have moderately increased, while retail emissions were noted as reducing due the lower carbon intensity of the grid.

Table 15 also shows the total emissions associated with household food waste from elements of the food value chain, and the percentage of the emissions from each element compared to the total emissions from household food waste. This shows that both home refrigeration and home cooking contribute under 5% of all emissions from household food waste in 2022 suggesting that reductions in the emissions from these two elements partly explain the overall reduction in GHG emissions from household food waste. The emissions associated with Distribution to retail, Retail and transport to home combined make up 18% of all emissions from household food waste. However as already noted, the emissions from this portion of the food value chain have remained similar in 2022 compared to 2012.

Overall, the decarbonisation of the energy grid is likely to contribute partly to the lower GHG emissions associated with household food waste in 2022 compared to 2012. However, it is not possible to assess the impact of any reductions in GHG emissions from food manufacturing which may or may not help to further explain reductions in tCO_2e in 2021.

Table 15: The emissions factors and total emissions associated household food waste split into elements of the food value chain.

	Element of the food value chain		
	Distribution to retail, Retail and transport to home	Home refrigeration	Home cooking
2012 emissions factor (tCO_2e/t)	0.70	0.31	0.62

2021 emissions factor (tCO_2e/t)	0.68	0.26	0.29
Percentage difference in emissions factor	-3%	-14%	-53%
2021/22 total emissions (tCO_2e)	3,200,000	710,000	710,000
Percentage of total FW emissions	18%	3.9%	3.9%

Disposal emission factors

The emissions factor associated with the disposal of food waste is 88% lower in 2022 compared to 2012 (Table 16). Subsequently the emissions related to the disposal of food waste in 2022 are 350,000 tCO2e compared to 3,200,000 tCO2e in 2012. In 2012, a larger proportion of food waste was sent to landfill which is associated with more emissions than disposal routes more commonly used in 2022 (e.g. anaerobic digestion, composting and incineration). The very low emissions for recycling destinations such as anaerobic digestion is partly an artefact of how these emissions are treated in a system analysis, with the emissions from recycling processes being attributed to the *new product* (e.g. in fertiliser production, or electricity generation) rather than the disposal of food waste. This is discussed further in the forthcoming report *UK Food System GHG Emissions*[47]. Therefore, changes in how food waste is disposed also contributes to the lower emissions associated with household food waste in 2022 compared to 2012.

Table 16: Disposal emission factors, total disposal emissions associated with household food waste and the percentage of the total edible food waste emissions from household food waste attributed to disposal for 2012 and 2022.

	2012	2021/22
Emissions factor (tCO_2e/t)	0.67	0.08
Emissions (tCO_2e)	3,200,000	350,000
Percentage of total FW emissions	13%	2.2%

[47] UK Food System GHG Emissions, WRAP, 2021, Accessed via: UK Food System GHG Emissions | WRAP

Conclusions: comparison of GHG emissions from 2021/22 with 2012 household food waste data

The emissions associated with food waste in 2012 and 2022 should not be directly compared because of differences in the scope of what the total emissions estimates include. In 2012, indirect land use is included in the estimates whereas direct land use is used in 2022 estimates. This means that 2012 estimates include emissions from a larger scope of factors contributing to why the 2012 estimate is larger than the 2022 estimate. Further work will be undertaken to understand the land use change emissions associated with household food waste in 2022.

The emissions estimate for 2012 is also likely to be greater than the estimate for 2022 partly because of reductions in CO_2e emissions in aspects of the food value chain. Specifically, the decarbonisation of the electricity grid is likely to have reduced emissions in food manufacture, retail and households. Additionally, changes in how food waste is disposed also contributes to the lower emissions associated with household food waste in 2022 compared to 2012.

Comparison with GHG emissions in the *UK Food System GHG* report

The emissions of household food waste have also been estimated in the *UK Food System GHG*[48] report which reports 23 million tonnes of CO_2e in 2019. The figures in this report should not be directly compared as a different methodology is used the *UK Food System GHG* report. In calculations from the *UK Food System GHG* report a top-down approach is used rather than a bottom-up approach used for the 2021/22 estimates. In a top-down approach, total emissions from stages of food value chain, such as total emissions from agriculture, are allocated to food waste. In a bottom approach, as used in the 2022 estimates, GHG emission factors are assigned to each food group and scaled up according to the weight of wasted food from each group. Therefore, in a bottom-up approach the composition of wasted food is used as a basis for understanding total GHG emissions.

In the *UK Food System GHG estimate* recent detailed food waste composition data was not available. Instead to understand the proportion of different types of food, the composition of food purchased is used in the calculations, which varies from the composition of wasted food. For example, when looking at the proportion of all purchases by food type, dairy & eggs, drinks and confectionary & snacks are bought in higher proportions compared to the proportion they are wasted in. These food types are associated with relatively high GHG emissions which may

[48] UK Food System GHG Emissions, WRAP, 2021, Accessed via: UK Food System GHG Emissions | WRAP

contribute to the higher emissions estimates in the UK Food System GHG report compared to those calculated in this research.

9.0 Uncertainty

9.1 Uncertainties associated with Local authority collected household food waste in the UK 2022 Compositional Data

Sampling of households within local authority compositional analyses

Random and systematic uncertainties

It is not practically feasible or desirable to perform compositional analysis on all households within a local authority, so samples were drawn. Random errors in the results from each compositional audit contribute to the variability observed among the audits, which in turn influences the confidence intervals presented in this report. If households are not selected randomly (which is often the case for practical reasons), there could also be a systematic error if there is a bias in the selection of households. This uncertainty has not been accounted for in this report and – should it exist – would be very difficult to quantify. The same sampling issues are also applicable to Household Waste Recycling Centres (HWRCs) and contamination of dry recycling by food waste. In both cases, the random, sampling errors have been estimated and reported.

Having data from only a sample of local authorities

Random uncertainties

Having data from a sample of local authorities (rather than the entire population) introduces random sampling error into the calculations. The degree of sampling error has been quantified in this study using confidence intervals, which take account of both the genuine variation in waste composition between local authorities and the additional variation caused by sampling households within local authorities (see previous uncertainty).

Representativeness of local authorities covered by compositional analysis

Systematic uncertainties

There are many factors that may influence whether a local authority performs a compositional analysis. Analysis of the data and findings in previous Local authority collected household food waste in the UK work suggests that local authorities who collect food waste separately had

lower food waste yields in the residual stream. This has been adjusted for in the calculations by stratifying the sample between those collecting food waste in targeted collection and those not.

Total quantity of waste in relevant streams (reported in WasteDataFlow)

Systematic and random uncertainties

Information reported to WasteDataFlow undergoes many checks before it is made public. However, it is possible that measurement, classification and reporting errors do occur, which would feed through to the results in this report. It is not possible to quantify the magnitude of these potential errors.

Estimate of food waste in mixed organics collections

Systematic uncertainty

The proportion of food waste in mixed organics collections is highly variable and depends on a range of factors including how the collection is communicated to households and the length of time since its introduction. This data is only available for a few authorities so it was necessary to make a generic assumption for the other authorities with such collections, which could have introduced a systematic error into the results. To test the sensitivity of the result to the yield of food waste assumed to be in mixed organics collections, yields were doubled and the results compared. The influence of this (relatively extreme) change was 2.6% for estimate of local-authority-collected waste in the UK.

Estimate of food waste in separate food waste collections

Systematic uncertainty

The analysis assumed that 100% of material in separate food waste collections is food or drink waste. Data from eleven studies suggests that contamination is low (average of 1.8%). Given this, the assumption made in the report is likely to result in an over-estimation of food waste of about 13,347 tonnes (0.3% of the total UK food waste tonnage). Additionally, a proportion of separate food waste collections are made up of food waste liners which are compostable and accepted by local authorities, but are not food waste. Of a separate food waste collection about 2.4% are liners, this equates to about 18,377 tonnes of lines in the UK per year. Subsequently the separate food waste estimate including liners means that the total UK food waste estimate is likely an overestimate by about 0.4% as result of liners being included in the reported tonnages.

9.2 Uncertainties associated with detailed waste compositional analysis

Sampling of households

Random and systematic uncertainties

It is not practically feasible to perform compositional analysis on all households in the UK, so a sample was drawn, leading to random sampling errors which are accounted for at the nation level in the confidence intervals quoted in this report. Sampling can lead to systematic errors if the sample of households is not representative of the population. Weighting was therefore applied to account for over- and underrepresentation of particular categories of household. However, some systematic uncertainty may remain from participation bias – those households agreeing to take part in the study may differ, with regard to food waste, to those that do not agree.

Measurement of waste

Random and systematic uncertainties

There is the possibility that uncertainties were introduced during the analysis phase. Calibrated scales were used to weigh items with a precision of 1 gramme and checks were carried out to ensure that the data was entered correctly in the electronic databases. Furthermore, unusual coding and weights were investigated during the analysis phase and any anomalies corrected. Therefore, it is likely that the risk of substantial uncertainty from this source is small.

Sampling from England and Wales only

Systematic uncertainty

The households sampled were in England or Wales but chosen to be representative of UK households on a number of variables. However, if there are systematic differences in the types of food and drink waste generated between England and Wales and the rest of the UK, this would bias the results. Previous research has indicated that differences in the total amount of waste are similar between nations, but little is known about differences in the type of waste.

Omission of flats

The sample for the detailed waste compositional analysis contained very few flats. This was because the research needed to identify which waste receptacles came from which properties,

allowing a link to be made between waste and household questionnaire responses. This omission is consistent with previous research and was explored in more detail in 2012[49].

9.3 Uncertainties associated with diary estimates

Sampling of households

Random and systematic uncertainties

It is not practically feasible to include all UK households in diary-based research, so a sample was drawn, leading to random sampling errors which are accounted for at the nation level in the confidence intervals quoted in this report. Sampling can lead to systematic errors if the sample of households is not representative of the population. Weighting was therefore applied to account for over- and under representation of particular categories of household. However, some systematic uncertainty may remain from participation bias – those households agreeing to take part in the study may differ, with regard to food waste, to those that do not agree.

Converting diary entries into weights

Systematic uncertainty

Not all items in the kitchen diaries were recorded with a weight: some were provided with a volume and others the number of items. Conversion of all items was made to weight, which introduces uncertainties which are difficult to quantify. It is likely, however, that over- and under-estimates of different food types will balance out to some extent and so reduce the extent of any bias.

Underestimation in Kitchen-diary 2022/23 research

Systematic uncertainty

Underestimation is a potential issue in all diary reporting. Analysis conducted in this project suggests that diaries showed a greater degree of underestimation than in previous WRAP diary studies. Subsequently diary data has been adjusted to account for this under estimation and is also not used to make year on year comparisons. This topic is discussed in detail in Chapter 7.0.

[49] See section 12.0 Investigation into the impact of omission of flats from the detailed waste compositional analysis in Methods used for Household Food and Drink Waste in the UK 2012, WRAP, 2013, Accessed via: Household Food and Drink Waste in the United Kingdom 2012 | WRAP

Non-response

Systematic uncertainty

The diary research could have been subject to various types of non-response bias: for example, people not agreeing to take part in the research, or people not completing the research. These would bias the results if they correlated with food waste arisings. As quota sampling was performed, it is hard to assess the impact on these systematic errors.

Sampling from England and Wales only

Systematic uncertainty

The households sampled for the diary were in England or Wales but chosen to be representative of UK households on a number of variables. However, if there are differences in the amount, types and disposal routes of food and drink waste between England and Wales and the rest of the UK, this would impact on the results. Previous research has indicated that differences in the total amount of waste are similar between nations, but little is known about differences in the type of waste.

9.4 Uncertainties associated with overall calculations

Number of Households in the UK

Systematic uncertainty

Household estimates are from the Office for National Statistics Families and households dataset.[50] Census data is viewed as being the most accurate estimate of households. As the census has not been used in this case, it is important to consider that any uncertainty will have most impact on the estimates derived directly from diary-based research (waste going down the sewer and home composted) because the average amount of waste per household is calculated and then multiplied by the number of households. This does not affect the results for the total amount of waste in the Local authority collected household food waste in the UK 2022 report using the 'standard' method, but does affect results expressed 'per household' and has a very minor influence on the 'alternative' method.

[50] Families and households - Office for National Statistics

Processing errors

Random and systematic uncertainties

It is possible that processing errors have occurred, including (but not limited to) errors in populating databases, errors in calculations and miscoding of items. Checks have been performed at all stages of the calculations to minimise the risk of these errors.

Seasonality and application of seasonality factors

Random uncertainty

To remove a systematic bias in the estimates, adjustment has been made for seasonality. The seasonality factors were derived from the Family Food survey, which is based on diary-based research subject to random, sampling uncertainties. This means that the application of these factors will introduce a small uncertainty into the calculations. Due to the complexity of including these uncertainties in the calculation of confidence intervals and the small impact on the overall uncertainty, these have not been incorporated in the overall uncertainty calculations. This means that the confidence intervals slightly underestimate the uncertainty for food types.

Application of cost and carbon factors

Random and systematic uncertainties

The carbon and cost factors used to determine the price of purchases that are wasted and the associated greenhouse gas emissions will be subject to a number of uncertainties. Some of these will relate to the primary research which they are based on (e.g. assumptions in life-cycle analyses; sampling errors in diary research used to determine food expenditure). In addition, some of the factors will be calculated for food and drinks that are not exact matches for the waste. Both these factors introduce uncertainties to the cost and carbon estimates above and beyond those relating to the calculation of the weight of waste. The uncertainties associated with these factors are difficult to quantify.

9.5 Confidence intervals

Calculation of confidence intervals

Confidence intervals have been calculated for key estimates in the report. This gives an indication of how precise the estimates are. The confidence intervals constructed only measure the effect of sampling errors. They do not include sources of systematic uncertainties as discussed in sections 9.2 and 9.3. In some cases, systematic errors have been explored; in the case of seasonal variation in food waste, adjustment has been made to account for it. However, for all other systematic errors the uncertainties have not been incorporated into the total uncertainty estimates. For this reason, confidence intervals stated are underestimates and represent a lower bound on the amount of uncertainty. Therefore, they are referred to as approximate confidence intervals.

For each of the primary data sources used in the report, approximate 95% confidence intervals were calculated. For the Local authority collected household food waste in the UK 2022 research, the variability between authorities was used to obtain the confidence intervals. For the detailed waste compositional analysis and diary-based research, the variability in the amount of waste between households was used. Although the distribution of the amount of waste from individual households is non-normal – the distribution is skewed with a right-hand tail – the error structure around the mean was assumed normal, invoking the central limit theorem. This approach should be valid given the number of households in each sample group (>30). The amount of UK household food and drink waste generated in 2021 was 6.0million tonnes ± 0.28 million tonnes (Table 17).

Table 17: Estimate and uncertainties of household food and drink waste by disposal route, rounded to two significant figures.

	2021-2023	
Disposal route	Millions tonnes/year	Confidence interval
LA collected	5.0	0.13
Sewer	0.81	0.24
Home composted	0.16	0.046
Total	6.0	0.28

Confidence intervals by food type

For calculating the confidence interval for a food type, an analogous approach was adopted to that described above. If the confidence interval was less than ±40% of the estimate, then it was included in this report. See Table 18 for the estimate of uncertainty of each food type according to the category 2 level.

Using a relative confidence level of ±40% as a cut off results in a fair degree of confidence for all food types reported. For the food types contributing a high proportion to the total food and drink waste in the UK (e.g. bread, potatoes, and meals), the relative confidence levels are around ±9 to ±10%, indicating a higher degree of certainty around the estimate. For those food types that just pass this cut off (i.e. the relative confidence level is just under ±40%), the estimate presented is a good indication of the level of waste, but not a precise estimate.

Table 18: Estimate of uncertainty around estimate of food type (% of estimate)

Category 1	Category 2	95% confidence interval
J - Drink	J.1 - Tea	6%
E - Fresh fruit	E.2 - Banana	8%
C - Dairy and eggs	C.6 - Egg	8%
G - Fresh vegetables and salads	G.15 - Potato (all chips in processed)	9%
A - Bakery	A.3 - Standard bread	9%
B - Meat and fish	B.5 - Poultry (chicken / turkey / duck)	9%
E - Fresh fruit	E.1 - Apple	10%
G - Fresh vegetables and salads	G.10 - Lettuce & leafy salad, including spinach	10%
O - Meals (home-made and pre-prepared)	O.2 - Composite meal	10%
B - Meat and fish	B.1 - Pork / ham / bacon	11%
G - Fresh vegetables and salads	G.14 - Pepper	11%
G - Fresh vegetables and salads	G.8 - Cucumber	11%

G - Fresh vegetables and salads	G.5 - Carrot	11%
G - Fresh vegetables and salads	G.13 - Onion	11%
G - Fresh vegetables and salads	G.2 - Broccoli	12%
E - Fresh fruit	E.13 Tangerine/satsuma/clementine	12%
A - Bakery	A.2 - Speciality bread	12%
E - Fresh fruit	E.5 - Orange	12%
G - Fresh vegetables and salads	G.18 - Tomato	12%
E - Fresh fruit	E.6 - Other citrus	12%
M - Cake and desserts	M.1 - All cakes and desserts	13%
C - Dairy and eggs	C.2 - Cheese	13%
I - Confectionery and snacks	I.2 - Savoury snacks	14%
E - Fresh fruit	E.3 - Grapes	14%
G - Fresh vegetables and salads	G.11 - Mixed vegetables	14%
O - Meals (home-made and pre-prepared)	O.3 - Sandwich	14%
G - Fresh vegetables and salads	G.20 - Other root vegetables	15%
G - Fresh vegetables and salads	G.6 - Cauliflower	15%
K - Condiments, sauces, herbs & spices	K.3 - All other condiments, sauces, herbs & spices	15%
G - Fresh vegetables and salads	G.3 - Cabbage (except Kale)	15%
G - Fresh vegetables and salads	G.12 - Mushroom	16%

G - Fresh vegetables and salads	G.19 - Other vegetables and salad	16%
G - Fresh vegetables and salads	G.9 - Leek	16%
O - Meals (home-made and pre-prepared)	O.4 - Savoury products	17%
E - Fresh fruit	E.10 - Avocado	18%
E - Fresh fruit	E.4 - Melon	18%
A - Bakery	A.1 - Morning goods	18%
B - Meat and fish	B.2 - Beef	18%
N - Other	N.3 - Mixed food from multiple categories	18%
I - Confectionery and snacks	I.1 - Chocolate and sweets	19%
G - Fresh vegetables and salads	G.16 - Spring onion	19%
N - Other	N.1 - Gunge	19%
E - Fresh fruit	E.11 - Stone fruit (excludes avocado)	20%
G - Fresh vegetables and salads	G.17 - Sweetcorn / corn on the cob	21%
A - Bakery	A.4 - Other bakery	21%
C - Dairy and eggs	C.1 - Milk	21%
J - Drink	J.2 - Coffee	22%
B - Meat and fish	B.6 - Other inedible meat and fish (bone, grissle, etc.)	23%
B - Meat and fish	B.3 - Fish and shellfish	24%
B - Meat and fish	B.4 - Lamb	24%
C - Dairy and eggs	C.4 - Yoghurt / yoghurt drink	24%
D - Staple foods	D.2 - Pasta	25%
E - Fresh fruit	E.7 - Pear	25%
D - Staple foods	D.3 - Rice	25%

E - Fresh fruit	E.8 - Pineapple	26%
K - Condiments, sauces, herbs & spices	K.2 - Cook in sauces	26%
D - Staple foods	D.1 - Breakfast cereal	26%
H - Processed vegetables and salad	H.8 - Potato - other processed potato products	26%
E - Fresh fruit	E.12 - Other fruit	27%
J - Drink	J.4 - Fruit juice and smoothies	28%
H - Processed vegetables and salad	H.2 - Other vegetables and salad	28%
H - Processed vegetables and salad	H.1 - Potato (including all chips)	30%
H - Processed vegetables and salad	H.5 - Hummus	30%
J - Drink	J.6 - Carbonated soft drink	30%
G - Fresh vegetables and salads	G.1 - Bean (all varieties)	31%
H - Processed vegetables and salad	H.6 - Baked beans	31%
I - Confectionery and snacks	I.4 - Other confectionery and snacks	31%
O - Meals (home-made and pre-prepared)	O.1 - Soup	31%
C - Dairy and eggs	C.3 - Cream and creme fraiche	32%
J - Drink	J.7 - Squash	36%
L - Oil and fat	L.2 - Fat	37%
H - Processed vegetables and salad	H.4 - Coleslaw	37%
K - Condiments, sauces, herbs & spices	K.1 - Gravy	37%

G - Fresh vegetables and salads	G.7 - Courgette	38%
B - Meat and fish	B.7 - Other edible meat and fish (unidentifiable / mixed meat / offal)	39%

10.0 Appendix

10.1 Questionnaire for participants of the Waste Composition Analysis

Screener question:

Q1 **How responsible are you for food shopping in your home?**

Q2 **How responsible are you for the preparation and cooking of food in your home?**

	Q1	Q2
I have responsibility for all or most of it	1	1
I have responsibility for about half of it	2	2
I have responsibility for less than half of it	3	3
I'm not responsible for any of it	4	4

IF RESPONSE IS 4 FOR EITHER Q1 OR Q2, THEN ASK IF THERE IS SOMEONE IN THE HOUSEHOLD WITH MORE RESPONSIBILITY FOR THESE ACTIVITIES AND INTERVIEW THAT PERSON. IF NO, CLOSE INTERVIEW.

Q3	Please read the following statements that people have made about food. Can you tell us to what extent you agree or disagree with each?					
		Definitely agree	Tend to agree	Neither agree nor disagree	Tend to disagree	Definitely disagree
1.	For me, food is just fuel to live	1	2	3	4	5
2.	The price of food doesn't really matter as long as I know that the quality is good	1	2	3	4	5
3.	I enjoy cooking and preparing food	1	2	3	4	5
4.	I often feel under time pressure in my day-to-day life	1	2	3	4	5
5.	Where food has come from and how it has been grown are very important to me	1	2	3	4	5
6.	Price is the main consideration when I'm shopping for food	1	2	3	4	5
7.	I often eat out and/or order takeaways	1	2	3	4	5

8.	It's a financial struggle for me to buy enough food	1	2	3	4	5

Q4	In the past month, how often have you done each of the following?				
	Eaten a meal in a canteen or staff restaurant	0 (none)	1-2 times	3-4 times	5+ times
	Eaten a meal in a restaurant, fast food restaurant, café, pub or similar	0 (none)	1-2 times	3-4 times	5+ times
	Purchased/ordered take away food to eat at home	0 (none)	1-2 times	3-4 times	5+ times
	Purchased food for 'on the go' (e.g. Pret a Manger, Eat, Boots, M&S, Tesco, etc.)	0 (none)	1-2 times	3-4 times	5+ times

Q5 In the last month, approximately how many times have you (or a member of your household) been, in-person to supermarkets and other grocery stores?

- 0
- 1 to 2
- 3 to 5
- 6 to 10
- More than 10

Q6 In the last month, approximately how many times have you (or a member of your household) had either on-line deliveries or used a 'click and collect' service for groceries?

- 0
- 1 to 2
- 3 to 5
- 6 to 10
- More than 10

Q7 **Which of the following have you used in the <u>past month</u> to (either in store or online)?**

Aldi	1
Asda	2
The Cooperative (Co-op)	3
Iceland	4
Lidl	5
Marks & Spencer	6
Morrison's	7
Ocado	8
Sainsbury's	9
Tesco	10
Waitrose	11
Ready-made meal kit/ recipe box (e.g. Hello Fresh)	12
Home delivery vegetable/ fruit box scheme (e.g. Abel & Cole)	13

Food banks or other food charities	14
Other	15

Q8 **Which of the following type of fridge(s) do you have and use at home? Please select all that apply**

Large capacity fridge (e.g. American-style fridge/freezer)	1
Medium capacity fridge (e.g. integrated fridge freezer)	2
Smaller capacity fridge (e.g. small under-counter fridge)	3
I do not own a fridge [SINGLE CODE]	4

Q9 **Which of the following type of freezer(s) do you have and use at home? Please select all that apply**

Large capacity freezer (e.g. large standalone freezer)	1
Medium capacity freezer e.g., integrated freezer	2
Smaller capacity freezer (e.g. two or three freezer trays)	3
A small freezer compartment in a fridge	4
I do not own a freezer [SINGLE CODE]	5

To what extent do you decide what you are going to eat for main meals in advance? Please just read out the letter that applies.

- A - I know what almost all of the main meals will be for the next week
- B - I know what most of the main meals will be for the next week
- C - I know what a few of the main meals will be for the next week
- D - I usually decide on the day

When you last purchased Apples, where did you store them?
When you last purchased Potatoes, where did you store them?

Q11 I'd like you to think about the last time you purchased fresh apples. After you returned home / after your delivery arrived, where did you store them? Please just read out the letters that apply.

A - In a fruit bowl, on a shelf or a worktop
B - In a cupboard
C - In the fridge
D - Other (do not specify)
E - Don't buy / store fresh apples
F - Don't know / can't remember

Q12 I'd now like you to think about the last time you purchased fresh potatoes. After you returned home / after your delivery arrived, where did you store them? Please just read out the letters that apply.

- A - In a cupboard or other dark place / in the dark
- B - On worktop / shelf / in pantry
- C - In fridge
- D - Other (do not specify)
- E - Don't buy / store fresh potatoes

- F - Don't know / can't remember

Q13 Please think back to the last time you cooked some rice. Did you measure the amount of rice used before cooking?

- Yes
- No
- Don't know / can't remember
- I don't cook rice

Q14 IF YES to above question: And what did you use to measure the rice?

- A measuring jug / cup
- Weighing scales
- Indicators on the side of the rice packet
- My own judgement
- Something else
- Don't know / can't remember

Q15	How often, if at all, do you have the following kinds of leftovers?					
		Regularly	Fairly often	Occasionally	Rarely	Never
A	A takeaway meal that you didn't finish	1	2	3	4	5
B	Food cooked or prepared at home and served on the plate, which you or your family didn't finish	1	2	3	4	5
C	Food cooked or prepared at home that wasn't served (e.g., more food was cooked than was needed)	1	2	3	4	5

Q16	For these different types of leftovers that arise, how likely are they to be eaten?				
		Always eaten	Usually eaten	Sometimes eaten	Never eaten
A	A takeaway meal that you didn't finish	1	2	3	4
B	Food cooked or prepared at home and served on the plate, which you or your family didn't finish	1	2	3	4
C	Food cooked or prepared at home that wasn't served (e.g., more food was cooked than was needed)	1	2	3	4

Q17 What do you understand the term USE BY date to mean on a food product, irrespective of what you would personally do with this information? Please read out the letter that applies.

- Last day on which food is at its highest quality
- Last day on which the food is safe to eat/use
- The day by which the food must be sold
- It depends on the product
- Not sure

Q18 What do you understand the term BEST BEFORE date to mean on a food product, irrespective of what you would personally do with this information? Please read out the letter that applies.

- Last day on which food is at its highest quality
- Last day on which the food is safe to eat/use
- The day by which the food must be sold
- It depends on the product
- Not sure

Q19	**Please choose one of the two answers shown for each of the following statements to complete the sentence correctly - to the best of your knowledge.** **Options = statement 1, statement 2 or don't know**		
	Statement	1	2
A	Apples last longest if they are stored…	At room temperature	In the fridge
B	To keep food at its best your fridge should be set to a temperature...	Between 1 and 5 degrees Celsius	between 5 and 8 degrees Celsius
C	Meat and poultry such as chicken breasts and sausages….	Can be frozen up until the end of the use by date	Can only be frozen on the day of purchase
D	Potatoes that have bad bits or grown shoots…	Can be eaten if you remove the bad bits or shoots	Can't be eaten

Q20 **How good or not would you say you/your household are at each of the following?**

A. Making a meal plan for the week ahead
B. Making a detailed shopping list
C. Buying the right amount of food for your household's needs
D. [ASK IF THEY HAVE FRIDGE] Managing/ keeping track of food in the fridge
E. [ASK IF THEY HAVE FRIDGE] Checking/ changing the fridge temperature on a regular basis
F. [ASK IF THEY HAVE FREEZER] Freezing raw meat, bread and leftovers
G. [ASK IF THEY HAVE FREEZER] Defrosting items from the freezer
H. Making a meal by combining foods/random ingredients you happen to have
I. Portion sizing – accurately judging how much food you need to prepare
J. Using up leftovers
K. Batch cooking (i.e. making several portions to store for later)

Response options:

- Excellent
- Good
- Satisfactory
- Poor
- I don't generally do this.

Q21	Do you ever buy or eat any of these products? Please select all that apply.		
	Bread	1	
	Fresh milk	2	
	Potatoes (purchased fresh)	3	
	Fresh chicken	4	
	Apples	5	
	None of the above [SINGLE CODE]	6	

Q. 22 Thinking about the last time you bought the following foods, approximately what percentage of the amount you purchased ended up being uneaten and thrown away (whether to a compost bin, ordinary bin, council food waste collection, or down the sink)?

FOR ANY ITEMS Indicated as purchased in previous question –

- Bread (including crusts)
- Milk (including what you got rid of after it was poured/served, e.g. on cereal)
- Chicken (cooked or uncooked) including any chicken skin not eaten but not including bones
- Potatoes (cooked or uncooked) including uneaten skins/peelings
- Apples

0%
The whole amount that I bought **was used up and eaten**

100%
The whole amount that I bought **was uneaten and thrown away**

Q23 Thinking about when you have thrown food away, how is it disposed of in your home? Please just read out the letters which apply.

- A - In the general refuse bin
- B – Composted at home
- C - Council Food Waste Collection
- D - Down the sink / drain
- E - Feed to animals (e.g. pets, birds)
- F - Other (do not specify)
- G - Don't know

Q24 Which, if any, of the following prevents you from reducing the amount of food that your household throws away? Please just read out the letters which apply.

- A – I don't waste any / much food
- B - It takes too much time
- C - There are more important things to worry about
- D - The cost of the food I waste doesn't bother me
- E - I don't need to reduce food waste because I compost most of our food waste at home
- F - I don't need to reduce food waste because I use a local authority collection for most of my food waste
- G - I don't think throwing away food has a negative impact on the environment
- H - I don't want to compromise on the choice of fresh foods I have
- I - I can't help producing food waste because of other people in my household
- J - I don't know how I can reduce food waste
- K - I can't buy the right amount of food (for me / my household)
- L - None of the above are a barrier for me
- M- Other (please specify)

Q25 **Please indicate on the scale below how you make decisions about when to eat or throw away the following foods?**

		I rely entirely on the date given on the pack	I rely mostly on the date given on the pack	I rely on a mixture of the date and my own judgement	I rely mostly on my own judgement	I rely entirely on my own judgement	Don't know	I don't buy / eat this product
A	Fresh milk	1	2	3	4	5	6	7
B	Yoghurt	1	2	3	4	5	6	7
C	Fresh potatoes	1	2	3	4	5	6	7
D	Apples	1	2	3	4	5	6	7
E	Cucumber	1	2	3	4	5	6	7
F	Raw meat (e.g. chilled chicken breast, sausages)	1	2	3	4	5	6	7
G	Bread	1	2	3	4	5	6	7

Q26 **In the past year, not including this interview, have you seen or heard anything about the following?**

		Yes	No
A	The amount of food that is thrown away/ wasted	1	2
B	How to plan, buy, store, or prepare food to help reduce the amount that gets thrown away in your home	1	2

Q. **27 Have you seen any of these in the past year (show images)? Please select all that apply.**		
LFHW logo		
Little blue fridge		
Snowflake		
Chill the fridge out		
A-Z storage advice		
None of these [SINGLE CODE ONLY]		

Q28	**Thinking about your cooking and food preparation, how much are the descriptions below like you?**					
		Very much like me	Like me	A little like me	Not much like me	Not at all like me
1.	I usually cook from scratch	1	2	3	4	5
2.	Saving time is one of the most important things for me when it comes to preparing meals	1	2	3	4	5
3.	I'm comfortable not using up all the food I buy	1	2	3	4	5

Q29	**To what extent do you agree or disagree with the following statements?**						
		Strongly agree	Agree	Neither agree nor disagree	Disagree	Strongly disagree	Don't know
A	Food waste is an important national issue	1	2	3	4	5	6
B	Everyone, including me, has a responsibility to minimise the food they throw away	1	2	3	4	5	6
C	Minimising my food waste deserves to be one of my top priorities	1	2	3	4	5	6
D	I have been making more of an effort lately to reduce my food waste	1	2	3	4	5	6
E	I don't really think the amount of food that households throw away is an issue because supermarkets and bigger industries waste so much more than me	1	2	3	4	5	6

Q30	**How much is the description below like you?**						
		Very much like me	Like me	Somewhat like me	A little like me	Not much like me	Not at all like me
1.	I buy items of clothing that I don't expect to keep wearing for long	1	2	3	4	5	6
2.	I enjoy keeping up to date with the latest fashion trends	1	2	3	4	5	6
3.	Shopping for clothes is a chore that I try to avoid	1	2	3	4	5	6
4.	I am an early adopter when it comes to smart technology and gadgets	1	2	3	4	5	6
5.	Finding the cheapest price is more important to me than buying the top brands	1	2	3	4	5	6
6.	I buy products that are designed to last	1	2	3	4	5	6
7.	I buy products and services from ethical companies	1	2	3	4	5	6

Q31	**To what extent are the following issues important to you personally? Give a score from 0-10 (where 0 = this is not an issue that is important to me and 10 = this is an extremely important issue to me). If you haven't thought about this issue before then please select this answer.** INCLUDE AN OPTION FOR 'NOT THOUGHT ABOUT BEFORE'	
1.	Global poverty (e.g. famine, access to clean water, medicine)	0-10
2.	Local issues in my community	0-10
3.	Politics	0-10
4.	Plastic waste	0-10
5.	Litter / fly-tipping in your local area	0-10
6.	Climate change	0-10
7.	Loss of countryside and natural landscapes	0-10
8.	Extinction of species / loss of natural habitats and deforestation	0-10

Q32 Thinking about all the people in your household, including yourself, which of these best describes their eating habits in the home?

1. Yourself
2. Person 2 (RECORD AGE)
3. Person 3 (RECORD AGE)
4. Person 4 (RECORD AGE)
5. Person 5 (RECORD AGE)

6. Person 6 (RECORD AGE)
7. Person 7 (RECORD AGE)
8. Person 8 (RECORD AGE)
9. Person 9 (RECORD AGE)
10. Person 10 (RECORD AGE)

- A - Fussy eater(s)
- B - Stick to known and trusted meals
- C - Happy to eat whatever is prepared
- Not applicable
- Don't know

Q33	**Here are some statements people have made about themselves. Please select how much you agree or disagree with each one.**					
		Definitely agree	Tend to agree	Neither agree nor disagree	Tend to disagree	Definitely disagree
	Recycling is too much of a hassle to bother with	1	2	3	4	5
	I am prepared to make lifestyle compromises to benefit the environment	1	2	3	4	5
	Everyone has a responsibility to help towards cleaning up the environment	1	2	3	4	5
	It's only worth doing environmentally friendly things if they save you money	1	2	3	4	5
	The effects of climate change are too far in the future to really worry me	1	2	3	4	5
6	Being green is an alternative lifestyle, it not for the majority	1	2	3	4	5
7.	The so-called 'environmental crisis' facing humanity has been greatly exaggerated	1	2	3	4	5

Q34	**Which of the following – if any – apply to you.**		
	Have a food sensitivity or intolerance (e.g. lactose)	1	

	Have a clinically diagnosed food allergy (e.g. gluten, nuts)	2	
	Follow a vegetarian diet	3	
	Follow a vegan diet	4	
	Follow a pescatarian diet (vegetarian, but eat fish)	5	
	Currently following some other form of diet e.g. Paleo, Dukan, Atkins, intermittent fasting (write in)	6	
	None of these [SINGLE CODE ONLY]	7	

Q35	**Are you…?**		
	Male	1	
	Female	2	
	Other (write in)	3	
	Prefer not to say	4	

Q36	**What is your gross household income before tax?**		
	Less than £10,000	1	
	£10,000-£19,999	2	
	£20,000-£29,999	3	
	£30,000-£39,999	4	
	£40,000-£49,999	5	
	£50,000-£69,999	6	
	£70,000-£89,999	7	
	£90,000+	8	
	I'd prefer not to say	9	

Q37	**Which of the following best describes your living situation?**		
	I live on my own	1	
	My partner and I live together without any children	2	
	I live with my children with no partner	3	
	My partner and I live with our children	4	
	I live in a shared house / with friends	5	
	I live in student halls of residence with a shared kitchen	6	
	Living with parents/other family	7	
	Something else (write in)	8	

Q38 Including yourself, how many people live in this household? _____

Q41	**For each adult in the household, please indicate which of the following best describes your current working status.** Note number of adults in each.		
	Working full time - working 30 hours per week or more	[__]	
	Working part-time - working between 8 and 29 hours per week	[__]	

	Self-employed - working 30 hours per week or more	[__]	
	Self-employed - working between 8 and 29 hours per week	[__]	
	Not working but seeking work or temporarily unemployed or sick	[__]	
	Not working and not seeking work	[__]	
	Student	[__]	
	Retired on a state pension only	[__]	
	Retired with a private pension	[__]	
	House person, housewife, househusband, etc.	[__]	

Q 42 What is the highest educational level that you have achieved to date?

1. No formal education
2. Primary school
3. Secondary school, high school, NVQ levels 1 to 3, etc.
4. University degree or equivalent professional qualification, NVQ level 4, etc.
5. Higher university degree, doctorate, MBA, NVQ level 5, etc.
6. Still in full time education
7. Don't know
8. Prefer not to answer

10.2 Weights of food items used in diary estimates

Table 19 The assumed weight in grams of each food item specified by a participant in the diary research

Units/food item	Weight in grams
apple	182
apple_core	35
apple_peel	22
apple_slice	18
apple_waste	57
apricot_piece	9
artichoke	120

artichoke_stem	35
asparagus_end	3
asparagus_spear	16
aubergine	425
aubergine_top	53
avocade_waste	50
avocado	80
avocado_peel	50
baby_corn	11
baby_tomato	15
bacon	30
bacon rinds	6
bacon_fat	6
bacon_medallion	30
bacon_pack	300
bacon_rind	6
bacon_slice	30
bag of chips	25
bagel	90
bageuette_end	24
baguette	150
baguette_piece	20
baked_bean	1
baked_bean_tin	420
banana	181
banana_inch	26
banana_peel	66
banana_skin	66
banana_skin	66

beef_cube	32
beef_rump_steak_fat	18
beef_slice	23
beetroot_jar	567
bhaji_small	49
biscuit	10
biscuit_pack	200
blinis	28
blueberry	2
bowl	350
bread_crust	5
bread_loaf	800
bread_pudding_slice	53
bread_roll	50
bread_slice	40
broccoli_head	375
broccoli_piece	11
broccoli_stalk	110
brocolli_stalk	110
brown_sauce_bottle	450
brussel_sprout	30
brussel_sprout_peel	6
bun	50
burger	54
cabbage	1100
cabbage_core	53
cabbage_leaves	32
cake	1000
cake_slice	68

carrot	140
carrot_peel	23
carrot_serving	80
carrot_stick	21
carrot_top	18
cauliflower	625
cauliflower_floret	42
cauliflower_leaves	6
cauliflower_stalk	110
cauliflower_waste	200
celery_stick_end	5
cellery_stick	40
cheese on toast	58
cheese_block	400
cheese_slice	25
cherry_stone	1
cherry_tomato	15
chicken_bone	50
chicken_bone_inch	4
chicken_breast	150
chicken_burger	114
chicken_carcass	210
chicken_drumstick	120
chicken_drumstick_bone	15
chicken_leg	200
chicken_leg_bone	30
chicken_nugget	15
chicken_piece	12
chicken_satay_skewers	16

chicken_skin	43
chicken_tender	18
chicken_thigh_bone	15
chicken_thigh_skin	16
chicken_wing_bone	38
chilli	20
chilli_top	4
chip	13
chips	13
chips_kfc_small	51
chocolate	5
chocolate_button	1
chocolate_square	7
clementine	74
clementine_peel	44
clove_peel	1
cob	89
cocktail_sausage	8
coffee pod	9
coffee_bag	14
coffee_pod	9
coffee_round	13
corn_cob	89
corn_on_cob	125
cornflakes_box	500
courgette	210
courgette_end	16
cracker	10
cracker_packet	300

crisp	2
crisp_packet	25
crossaint	44
crumbs	2
crumpet	55
crust	5
crusts	5
cucumber	301
cucumber_cm	10
cucumber_end	5
cucumber_peel	23
cucumber_pickled	36
cucumber_slice	5
cup	240
dessert_spoon	12
duck_carcass	1000
egg	60
egg_shell	6
egg_white	40
egg_yolk	20
english_muffin	72
feta_piece	10
fish_and_chips	454
fish_bone	5
fish_cake	135
fish_finger	28
fish_skin	15
flapjack	40
flour_bag	1500

floz	30
fosters_can	440
frozen_fries_bag	900
frube	37
garlic_bread_piece	17
garlic_bread_slice	21
ginos_provola_flatbread	284
grape	7
grape_bunch	232
grape_punnet	500
grape_stalk	5
grapefruit	340
grapefruit_rind	75
grapes	7
green_bean	4
ham_fat	2
ham_slice	13
handful	120
hash_brown	47
hot_cross_bun	70
hot_cross_bun	70
jam_jar	400
kiwi	69
kiwi_peel	20
kiwi_skin	20
l	1000
lamb_bone	150
lamb_chop_bone	25
large_mcdonalds_fries	144

lasagne	400
laurel_leaf	0
leek	164
leek_cm	8
leek_end	80
leek_waste	92
lemon	120
lemon_peel	83
lemon_slice	12
lettuce	539
lettuce_leaf	8
lime	67
lime_peel	46
mandarin_peel	20
mango	205
mango_peel	29
mango_waste	68
meatball	28
melon_rind	270
melon_skin	270
mince_pack	500
mini_pork_pie	50
mini_weetabix	3
ml	1
mug	300
mushroom	23
mushroom_peel	1
mushroom_stalk	6
naan_bread	90

nectarine	80
nectarine_peel	20
onion	95
onion_peel	25
onion_ring	15
onion_skin	25
orange	190
orange_peel	55
pack_noodle	85
pan	2839
pancake	113
parsnip	150
parsnip_peel	23
pasta	2
pasta_piece	2
pasta_plate	117
peach_slice	25
peach_stone	39
pear	138
pear_core	24
pear_peel	22
pear_stalk	1
peas	0
pepper	135
pepper_waste	30
philadelphia_light_and_tasty	42
pineapple_waste	250
pint	568
pitta_bread	57

pizza_crust	8
pizza_slice	32
plate	240
plum_stone	3
popcorn_chicken	4
pork_chop	250
pork_chop_bone	30
pork_joint	2500
pork_loin_chop	250
pork_loin_chop_fat	50
pork_rib_bone	28
pork_slice	40
porridge_scoop	15
potato	235
potato_peel	21
potato_scone	30
potatoes_bag	2000
potoato_peel	21
prawn	3
prawn_cracker	2
pumpkin	7938
quiche	400
quiche	160
raisin	1
raspberries_pack	150
ready_meal	400
rib	78
runner_bean	26
salad_bag	100

salad_potato	60
salmon_skin	15
sandwich	185
satsuma	85
satsuma_peel	20
sausage	67
sausage_roll	130
sausage_slice	7
scone	70
shallot	50
spring_onion	15
spring_onion_heads	8
steak	255
steak_gristle	MISSING
steak_pie	200
stock_cube	10
strawberry	12
strawberry_jelly_pot	125
strawberry_top	1
swede	900
swede_peel	80
sweet	5
sweets_pack	150
tangerine	88
tangerine_peel	31
tango_can	330
tbspn	15
teabag	14
teacake	63

tin	400
tomato	75
tomato_slice	15
tomato_stalk	5
tomato_vine	5
tspn	5
tub	500
turkey_crown	2250
turnip	250
turnip_peel	22
waffle	25
weetabix	19
wrap	61
wrap_filled	185
wrap_inch	31
yoghurt	120
yorkshire_pudding	19

10.3 Environmental factor proxies

Table A1: Categories requiring a substitute factor and the proxy used. Ingredient average means that an average of the components ingredients as been used

Food category	Substitute factor source
A.1 - Morning goods	Brioche
A.4 - Other bakery	All bread average
B.6 - Other inedible meat and fish (bone, grissle, etc.)	Meat average

D.2 - Pasta	Ingredient average
E.13 Tangerine/satsuma/clementine	Orange and other citrus average
G.16 - Spring onion	Onion
G.4 - Kale	Cabbage
G.9 - Leek	Onions & Leeks
H.4 - Coleslaw	Ingredient average
H.5 - Hummus	Processed vegetables average
I.3 - Sweet biscuits & cereal bars	Ingredient average
I.4 - Other confectionery and snacks	Average of other confectionary and snacks
J.1 - Tea	Average of other drinks
J.10 - Other drink	Average of other drinks
J.7 - Squash	Fruit juice and smoothie
K.1 - Gravy	Average of all other cats
N.1 - Gunge	Average of all other cats
N.2 - Drainings from canned food	Average of any products that are canned
N.3 - Mixed food from multiple categories	Average of all other cats
N.4 - Other	Average of all other cats
O.1 - Soup	Average of all other cats
O.2 - Composite meal	Average of all other cats
O.2 - Composite meal	Average of all other cats
O.3 - Sandwich	Ingredient average
O.4 - Savoury products	Average of all other cats

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